

Link Dynamics and Welfare in Production Networks*

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Abstract

Production networks are dynamic: firms continually add, deepen, and sever supplier relationships. Using the universe of firm-to-firm transactions in Chile, I document three facts about relationship life cycles: (i) a small mass of long-tenured relationships accounts for a disproportionate share of network sales; (ii) relationship size rises sharply with tenure; and (iii) the hazard of relationship dissolution declines steeply with duration. These patterns imply strong selection forces and persistent heterogeneity at the relationship level. To interpret these facts and quantify their macroeconomic consequences, I develop a tractable general equilibrium model where the production network evolves endogenously through continuous link intensities. The model nests two leading primitives for relationship dynamics within a unified framework: 1) persistent match-specific productivity $r_{ij,t}$ that shifts the effectiveness of an input within the buyer's aggregator, and 2) persistent relationship costs $f_{ij,t}$ that govern the feasibility of maintaining connections, as commonly assumed in the literature. Although both primitives can generate high link turnover, they imply fundamentally different mechanisms for selection, within-relationship growth, and the evolution of the network. The analysis shows that

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only persistence in match-specific productivity can replicate the observed within-relationship patterns. More broadly, the choice of primitive governing relationship dynamics shapes which links survive and expand, altering equilibrium input price indices and generating downstream effects that private firms do not internalize, leading to different welfare and TFP dynamics even in otherwise identical economies.

I Introduction

Production networks are not static objects. The network of buyer-seller relationships reshapes continuously as firms discover, expand, and abandon customers. In fact, relationship turnover is large. Roughly half of buyer-seller relationships observed in a given year will not be active the next year.¹ These rates far exceed job destruction (10-12%) (Davis and Haltiwanger, 1990) and firm exit (about 20%) (Bartelsman et al., 2009), suggesting network relationship dynamics may be a first-order margin of adjustment rather than a residual of firm-level reallocation.

The process governing link reallocation matter for macro outcomes because a relationship is not merely a bilateral trade, it determines buyers' marginal costs, and thereby the prices faced by downstream customers. As a result, link formation and survival decisions propagate through the network, affecting welfare and the transmission of shocks.

In this paper, I study the determinants of the evolution of production network, analyzing how firms select their business partners and its implications for welfare and efficiency. I use the data covering the universe of firm-to-firm transactions in Chile between 2017 and 2024. One striking fact about link selection emerges. Although important in number (56%), newly formed bilateral relationships contribute with only 12% of the value of the network. This suggest that understanding the lifecycle of business relationship can shed light into the determinant of firm-to-firm production networks. Therefore, to understand the fundamental driver of network selection I exploit the lifecycle of business relationship.

Two facts help identify the fundamental process governing business relationship. First, the hazard rate of the business link decreases with age, even when controlling for unobserved buyer and seller characteristics, suggesting a dynamic exit decision process driven by a bilateral fundamental. Second, link size increases sharply with age, suggesting a bilateral fundamental with productive value driving the selection of relationships. Importantly, differences in size at the beginning of the links' lifecycle translate to differences in duration of the relationship, indicating persistence in the productive bilateral fundamental. Together, these facts imply that relationship-level selection and persistence play a central role: long-lived relationships are not simply those that avoid exit by chance; they are systematically larger and more stable. Understanding the forces behind this selection mechanism is essential for assessing the macroeconomic consequences of network formation.

¹In fact, using firm-to-firm data from Chile, we document that 51% of all active relationship in a given year will be terminated by the end of the year. Similar magnitudes are reported for Belgium (43%) (Dhyne et al., 2021) and Japan (50%) (Koike-Mori and Okumura, 2024)

These empirical findings motivate the inclusion of a bilateral component $r_{ij,t}$ with productive value to model endogenous network selection. The match-specific compatibility shifter $r_{ij,t}$ captures the suitability of the good sold by supplier j for the production process of buyer i in period t . The model features a static equilibrium with dynamic states, where firms' productivity and bilateral compatibility evolves following exogenous AR(1) processes.

The endogenous network formation model features an homogeneous household with CES preferences, and a continuum of differentiated monopolistically competitive varieties with a Cob-Douglas production function. In the model, firms have to pay for each buyer. Sellers optimal decision to form links balance the private profits generated by the relationship, with a common fixed maintenance cost. Therefore, labor provided by the household is rented by firms for production and for network maintenance. We can characterize the endogenous network decisions by a threshold on bilateral compatibility, that endogenously determine the probability of trade and the quality-adjusted sourcing capacity between any pair of firms.

The model is then estimated. The parameters governing firms productivity are structurally estimated, and the parameters governing bilateral compatibility and firms' network formation decision are estimated using simulated method of moments. The parameters replicate the empirical facts previously described, and the model is able to replicate the distribution of firm-to-firm sales, as well as the in-degree and out-degree gradients by firm size.

To study the aggregate implications of endogenous network adjustment, we analyze the response of the economy to exogenous firm productivity shocks. Importantly, welfare changes are no longer fully characterized by the cost-based Domar weight, as in the case of fixed network models ([Baqae and Farhi, 2019](#)). Network adjustment opens two new channels that affect welfare: changes in the use of labor (production versus network maintenance), and changes in the sourcing capacity of the firms. The first channel emerges because to sell in the production network, firms need to hire labor to maintain and operate its relationships with customers. Therefore, shocked firms and adjusts their networks and also change their allocation of labor. The second channel emerges because by adjusting its connections, the sourcing capacity of buyers change, making them relatively more or less productive.

Simulation results show that although the cost-based Domar weights are not sufficient to characterize welfare changes, the network margin of adjustment plays a secondary role. In fact, endogenous network adjustment only contributes with less than 1%. In the model, an increase in network density is traduced into an increase in the probability of trade. However, because already all firms are essentially trading with each other at some small probability, the change induced by productivity shock only

moves marginal links, which are of relatively low bilateral compatibility or firm productivity, hence contributing less.

Its important to highlight that in the model there are only 2 distortions. First, double marginalization, because firms have (constant) markups and operate in a supply chain. And second, there is a network externality, where

Importantly, the bilateral compatibility component allows to study network formation in the extensive and intensive margin

Notably, firms' decision to create and destroy links has two opposing forces. On one hand, making new connections with other firms

The literature has focused on models with fixed-cost to explain endogenous network decisions (Dhyne et al., 2021; Bernard et al., 2022; Lim, 2018). However,

These considerations raise a central question: Do different microfoundations for relationship dynamics lead to materially different predictions for network evolution, aggregate TFP, welfare, and shock propagation? The answer matters for how economists conduct counterfactuals in general, and particularly so in production-network models due to amplification effects. This project combines rich administrative data from Chile with a tractable general equilibrium model to study the aggregate implications of different selection mechanisms.

Using the universe of firm-to-firm transactions in Chile (2017-2024), I document three facts about relationship life cycles. First, a small mass of long-tenured relationships accounts for a disproportionate share of total network sales. Second, relationship size increases sharply with tenure, producing a steep tenure-size profile. Third, the hazard rate of relationship dissolution declines strongly with duration. Together, these facts imply that relationship-level selection and persistence play a central role: long-lived relationships are not simply those that avoid exit by chance; they are systematically larger and more stable. Understanding the forces behind this selection mechanism is essential for assessing the macroeconomic consequences of network evolution.

To interpret these patterns and quantify their aggregate implications, the paper develops a tractable general equilibrium model of production networks with monopolistic competition, CES final demand, and CES intermediate input aggregation.² The key departure is that the network evolves endogenously through continuous link intensities.

To quantify the aggregate effect of different selection mechanism, the model nests two dynamic primitives within a unified structure. Each potential relationship (i, j) is characterized by a pair of link-level states $(r_{ij,t}, f_{ij,t})$. Match-specific productivity

²The model resembles the structure of Bernard et al. (2022), with the inclusion of dynamic states.

$r_{ij,t}$ scales the effectiveness of seller j 's input in buyer i 's CES intermediate aggregator, thereby directly affecting the buyer's input price index, marginal cost, and equilibrium sales. Relationship cost $f_{ij,t}$ captures the resource cost of maintaining relationship intensity. Both processes evolve exogenously as AR(1) processes, so the economy's dynamics arise from the evolving cross-sectional distribution of link states and the induced evolution of the network.

The paper's key conceptual point is that the nature of persistence – whether it reflects match quality $r_{ij,t}$ or cost stickiness $f_{ij,t}$ – changes which relationships survive and expand, and therefore changes the path of the endogenous network. In the match-quality primitive, selection operates on fundamentals that directly affect marginal costs and prices; high-quality relationships become endogenously more important, generating persistent heterogeneity and gradual within-relationship evolution. In the cost primitive, continuation is driven by fluctuations in relationship costs; absent a match-quality channel, within-relationship dynamics are difficult to generate in a quantitatively meaningful way without imposing additional mechanisms, as indicated by simulation results.

The documented empirical facts help to discipline the economic mechanism behind relationship persistence. In particular, they distinguish between selection driven by persistent economic fundamentals (match quality or suitability) versus selection driven primarily by persistent relationship costs that govern continuation without altering within-relationship productivity. This distinction is macro-relevant. Because production networks amplify distortions and reallocations through input-output linkages, differences in which links survive and grow translate into different aggregate TFP and welfare dynamics. Moreover, the same difference affects the propagation of shocks: a network shaped by persistent match quality adjusts and transmits shocks differently from a network shaped by persistent cost shocks and turnover.

Another key conceptual point relates to the planner's problem and efficiency: private relationship choices do not internalize the downstream price-index effects of links that propagate through the network. In this environment, the welfare impact of increasing a single relationship intensity can be written as a downstream-weighted sum of price-index effects. This provides a transparent sufficient statistic for the social value of relationship intensity and clarifies why decentralized network evolution can be inefficient even absent firm entry/exit.

The paper makes four contributions.

First, it provides new evidence on relationship life cycles using universe-level transaction data from Chile. Long-tenured relationships account for a disproportionate share of sales, relationship size rises strongly with tenure, and dissolution hazards decline

steeply with duration. These facts are robust across sectors, firm size classes, and sampling restrictions.

Second, it develops a unified general equilibrium framework of endogenous production networks with continuous link intensities that nests two leading dynamic primitives-persistent match quality and persistent relationship costs-within the same CES production structure. This nesting allows disciplined comparisons of network evolution, TFP, welfare, and shock propagation across primitives without changing the rest of the economy.

Third, it formulates a planner problem and derives a sufficient-statistic representation for the welfare value of marginal relationship intensity as a downstream-weighted sum of price-index effects. This delivers a precise notion of the link wedge: decentralized relationship choices fail to internalize network-wide downstream effects, providing a microfounded inefficiency specific to production networks.

Fourth, the quantitative framework is going to be used to evaluate how modeling choices about relationship dynamics shape aggregate outcomes in production networks. Holding the static environment fixed, the framework enables counterfactual comparisons across alternative persistence primitives-match-specific productivity $r_{ij,t}$ versus relationship cost stickiness $f_{ij,t}$ -and provides a transparent mapping from link-level dynamics to welfare. This structure makes clear that relationship microfoundations are not an innocuous modeling choice in production-network general equilibrium: different sources of persistence imply distinct patterns of network evolution and, consequently, different welfare accounting and policy counterfactuals.

Our paper builds on the production-networks literature that provides the general-equilibrium workhorse for studying how input-output linkages shape aggregate outcomes. Seminal contributions such as [Acemoglu et al. \(2012\)](#), [Carvalho \(2014\)](#), and [Baqae and Farhi \(2019\)](#) establish a tractable CES-based framework in which equilibrium price indices, substitution patterns, and downstream linkages determine aggregate propagation. A unifying feature of this literature is that the production network itself is taken as a primitive and static object: the set of buyer-supplier relationships does not respond endogenously to shocks or incentives. Our contribution is to retain this general-equilibrium structure-CES demand and input aggregation, monopolistic competition, and price-index propagation-while endogenizing the evolution of the network in a general setting with dynamic states.

A second literature develops models of endogenous production-network formation. [Oberfield \(2018\)](#) provides a theory of equilibrium input-output architecture in which firms' sourcing decisions jointly determine network structure, emphasizing the role of complementarities and statistical network properties. More closely related to our

approach, [Lim \(2018\)](#) and [Bernard et al. \(2022\)](#) develop tractable general-equilibrium environments in which buyer-seller relationships are formed subject to fixed or sunk costs, generating sparse endogenous networks and extensive-margin adjustment. In these models, persistence in relationships arises primarily from cost-side frictions, while conditional on survival the economic value of a relationship is largely static. Our framework aligns closely with this tradition: we nest the fixed-cost mechanism studied by [Lim \(2018\)](#) and [Bernard et al. \(2022\)](#) as a special case, and extend it by allowing for persistent match-specific productivity that directly affects equilibrium prices and input demand. This extension enables a direct comparison between alternative microfoundedations of link persistence within the same general-equilibrium environment.

Finally, our analysis relates to work on efficiency, misallocation, and welfare in disaggregated economies. [Hsieh and Klenow \(2009\)](#) and [Baqaee and Farhi \(2019\)](#) provide general frameworks for measuring allocative efficiency and aggregate productivity in economies with heterogeneous firms and input-output linkages, typically treating the network structure as given. [Lim \(2018\)](#) also discusses how relationship frictions affect aggregate outcomes through network formation. Our contribution differs in focus: we show that different selection mechanisms governing relationship persistence—cost stickiness versus persistent match quality—imply different welfare and TFP conclusions even when the static production environment is held fixed. By formulating a planner problem, we highlight a link-level wedge arising from downstream price-index effects that private firms do not internalize, clarifying why the source of relationship persistence is central for welfare accounting in production-network general equilibrium models.

The remainder of the paper is organized as follows. Section [II](#) describes the data and documents the empirical facts motivating the model. Section [III](#) presents the general equilibrium framework, nests the dynamic primitives, and characterizes equilibrium. Section [IV](#) discuss preliminary simulations of the model targeting empirical moments. Section [V](#) formulates welfare, derives the planner problem, and establishes the link wedge and sufficient-statistic welfare expressions. Section [VII](#) concludes.

II Empirical Evidence

This section presents the empirical evidence that motivates the inclusion of a link-level fundamental shaping the life-cycle patterns. The main goal is to understand how selection operates at the link level.

II. A Data

The main dataset consists of firm-to-firm administrative records from Chile, covering all transactions between firms from 2017 to 2022.³ It includes anonymous firm identifiers for both the seller and buyer, which facilitates tracking transactions between them over time. The raw dataset contains more than 3 billion observations corresponding to every transaction between the seller and the buyer. To make the dataset easily manageable, it is then collapsed at the year level, with a resulting 86 million observations. The main variable of interest is the total value of all transactions for each specific seller-buyer pair within a given year.

The main datasource is matched with firms' tax records from the Chilean IRS to obtain information on the number of employees, total sales, and the economic sector of each firm. Only firms with positive sales and at least one worker remain in the final sample. Additionally, the baseline sample includes only transactions between firms in the private non-agricultural sector.

For the empirical analysis, we define the following terms. We define links or relationships as unique seller-buyer pairs. However, some relationships exhibit relatively long periods of inactivity without trade. We classify a seller-buyer pair as a new relationship if over 12 months have passed without trade between them. We define the life of a link as the maximum consecutive years of trade between the seller-buyer pair, and we define a link's tenure or age as the total number of years the relationship has been active, with the first year of the relationship considered as age or tenure 1. Finally, we define link size or value as the total amount traded (in pesos) in a given year between the seller-buyer pair. In terms of notation, when discussing the link (i, j) , this refers to the link where firm i serves as the seller and firm j as the buyer, therefore link $(i, j) \neq (j, i)$.

II. B Empirical Findings

***Fact 1:** Long-term links tend to contribute disproportionately to network sales.*

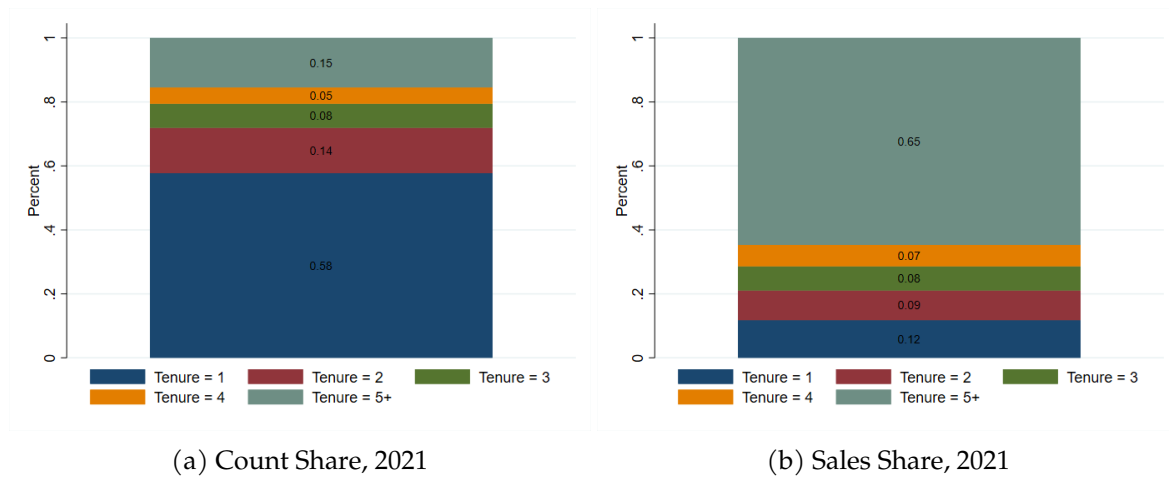
Figure 1 presents the distribution of the number of links in panel 1a and the distribution of link values or sizes in panel 1b. New links, defined as links with a tenure of 1 year,

³This study was developed within the scope of the research agenda conducted by the Central Bank of Chile (CBC) in economic and financial affairs of its competence. The CBC has access to anonymized information from various public and private entities by virtue of collaboration agreements signed with these institutions. To secure the privacy of workers and firms, the CBC mandates that the development, extraction, and publication of the results should not allow the identification, directly or indirectly, of natural or legal persons. Officials of the CBC processed the disaggregated data. All the analysis was implemented by the authors and did not involve nor compromise the Chilean IRS. The information contained in the databases of the Chilean IRS is of a tax nature originating in self-declarations of taxpayers presented to the Service; therefore, the veracity of the data is not the responsibility of the Service.

constitute 58% of the total number of relationships in 2021. In contrast, long-term links, defined as links with 3 or more years of age, account for only 28% of the total number of links in 2021, as shown in Figure 1a.

The picture depicted in figure 1b shows an opposite trend, with new links accounting for merely 12% of the total value, while long-term links represent 80% of the total network value in 2021. Figure 1 presents results for a specific year, but the validity of the conclusions extends beyond this selection. Additionally, Figure A1 in the appendix displays results excluding links for which we cannot identify the first year of the relationship due to data limitations.⁴

Figure 1: Distribution of Links



Note:

Therefore, most links are new, while most trade occurs between established relationships. This observation remains robust even when excluding the retail and wholesale sectors.

The fact that long-term links are relatively few but contribute disproportionately to the production networks' value suggests that the average size of long-term links should be significantly bigger. This insight leads us to our second fact.

Fact 2: *Positive association between the tenure and size of the link*

Figure 2 presents the average link size differential for links of various ages relative to a new link. To obtain the link size differential, I estimate the following equation and plot

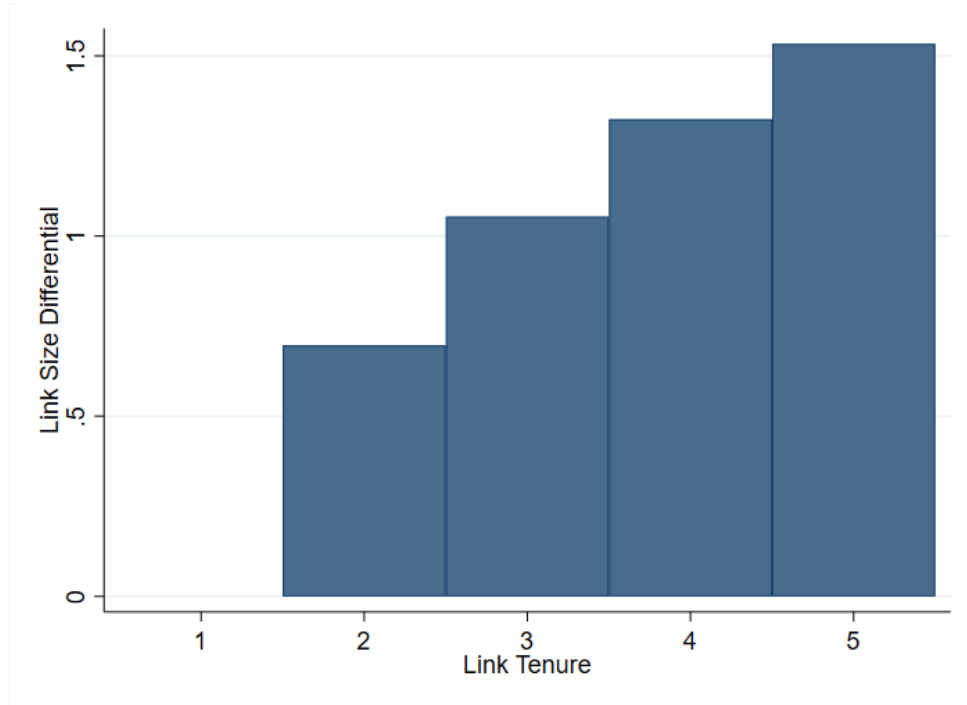
⁴Note that because the data start on 2017, we cannot identify the age or life of links that are active this year. Thus, Figure A1 in appendix account for this issue showing remarkably similar patterns than Figure 1.

the coefficients β_a

$$\ln S_{ijt} = \alpha + \sum_{a=1}^6 \beta_a \mathbb{1}\{age_{ijt} = a\} + \varphi_{it} + \vartheta_{jt} + \varepsilon_{ijt} \quad (1)$$

, where $\ln S_{ijt}$ represents the log value of the link between seller i and buyer j in year t , $\mathbb{1}\{age_{ijt} = a\}$ is an indicator variable for the age or tenure of the link between firms (i, j) in year t , and φ_{it} and ϑ_{jt} are seller-year and buyer-year fixed effects, respectively. I rely on the identification of the fixed effects based on each firm, buyer and seller, maintaining multiple links each year. The age or tenure of the link is identified only for links that started in 2018, the second year in the sample.

Figure 2: Tenure Profile



Note:

The coefficient β_a indicates the average difference in size between links of tenure a and links from year 1, which serves as the omitted category in the estimation. The result in Figure 2 demonstrates a positive association between link age and size, as previously suggested. In fact, links that are 5 years old are, on average, 7 times larger than new links. The positive association between link age and size is robust even when excluding the retail and wholesale sectors.

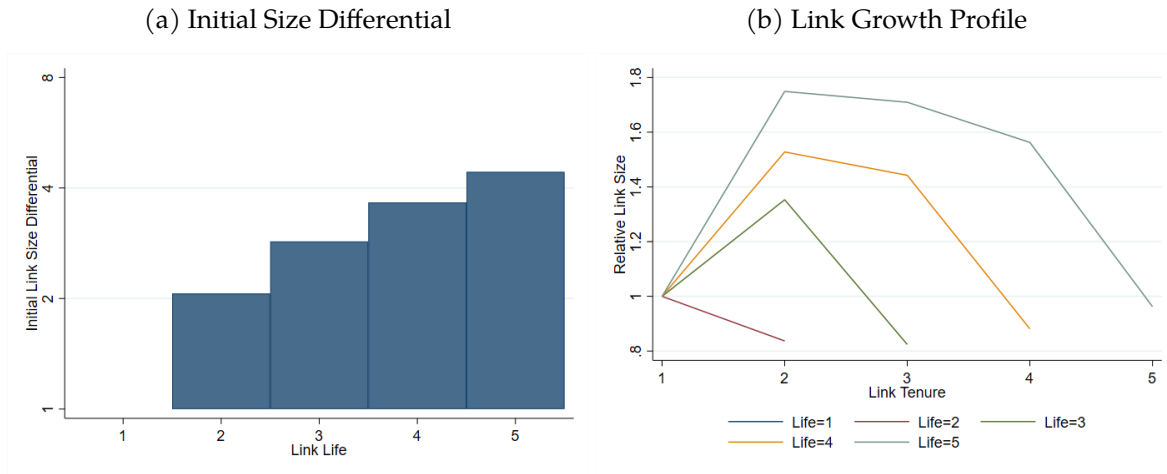
The significant size gap for long-term links may arise from two factors. First, long-term links might initially be larger, suggesting the selection of the best links over time. Second, long-term links could grow faster and, by definition, for a longer period. To

explore each case, equation 1 is augmented with an interaction term involving the link's life. Therefore, coefficients β_{al} provide an estimate of the size differential for a link at age a with a maximum duration of l relative to a new link.⁵ Results are presented in Figure 3.

$$\ln S_{ijt} = \alpha + \sum_{a=1}^6 \sum_{l=1}^6 \beta_{al} \mathbb{1}\{age_{ijt} = a\} \mathbb{1}\{life_{ij} = l\} + \varphi_{it} + \vartheta_{jt} + \varepsilon_{ijt} \quad (2)$$

Figure 3a presents the estimates of β_{1l} for $l = 1, 2, 3, 4, 5$, that is the size differential of links of varying lifespan during their first year of activity ($age = 1$). The results reveal significant differences in the first year of the links. Initially bigger links may live longer. In fact, links active for 5 years are, on average, 4 times larger in their first year than those that last only one year. This result suggests a relevant role for selection, which forces small, low-quality links to exit.

Figure 3: Link's Age-Life Profile



Note:

Figure 3b presents estimates of β_{al} normalized by the coefficient of the initial year β_{1l} for each lifespan l . The figure indicates that longer lifespans allow for more time to grow, rather than exhibiting significantly steeper growth trajectories. Furthermore, the observation that links shrink in the last year of their lifespan extends beyond this dataset. For example, [Monarch and Schmidt-Eisenlohr \(2023\)](#) find similar patterns in US import data.

The positive association between link size and duration indicates that larger links tend to persist for a longer time. This implies that long-term links first-order stochastically dominate (FOSD) short-tenured links. Figure A2 presents the cumulative distri-

⁵By definition, a new link is a link of age 1 and maximum duration or life equal to 1.

bution for links of different age. As previously discussed, as links mature they tend to be larger because they are initially bigger (selection), and because they had more time to grow. Thus, the distribution of long-term links is shifted to larger values compared to the distribution of younger links.

The fact that initially bigger links tend to live longer suggests a relationship between link age and survival. This insight leads us to our third fact.

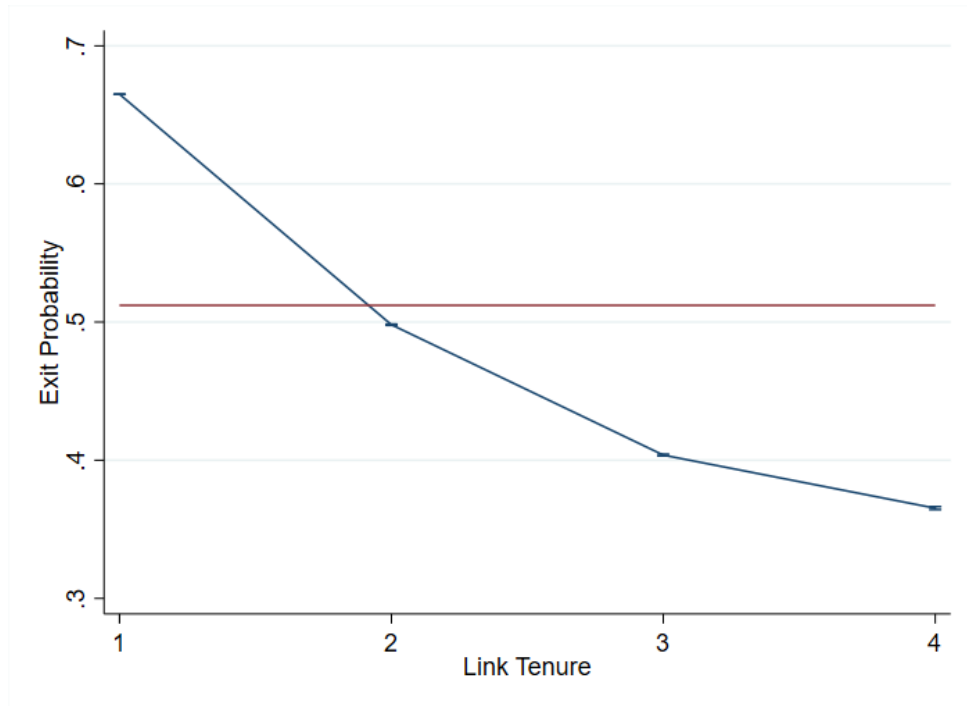
Fact 3: *Negative association between the hazard rate and the age of the link.*

Figure 4 presents the relationship between the hazard rate and link's age or tenure. To estimate the exit probability of a link, I estimated equation 3, where $\mathbb{1}\{Die_{ijt}\}$ is an indicator variable reflecting whether the link between firm (i, j) in year t will deactivate in year $t + 1$.

$$\mathbb{1}\{Die_{ijt}\} = \alpha + \sum_{a=1}^5 \gamma_a \mathbb{1}\{age_{ijt} = a\} + \varepsilon_{ijt} \quad (3)$$

The coefficient of interest is γ_a , which measures the probability that a link of age a terminates.⁶ Given that the data ends in 2022, I can only identify links that exited until 2021.

Figure 4: Hazard Rate



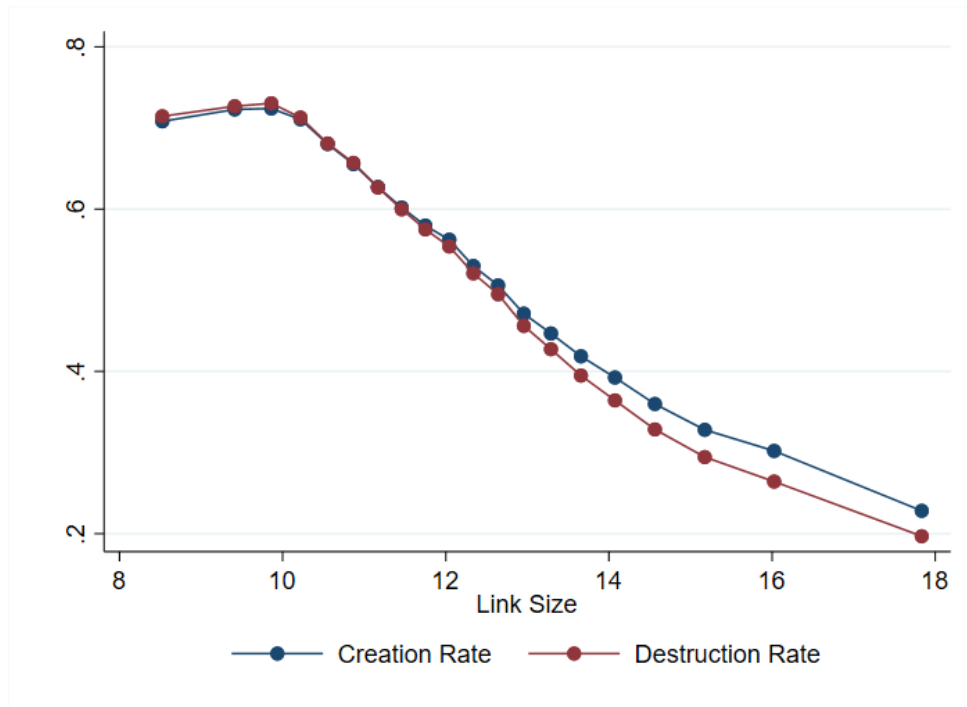
Note:

⁶In fact, because $\mathbb{1}\{age_{ijt} = a\}$ is an indicator variable as previously discussed, γ_a identifies the differential in the exit probability relative to the omitted category.

The results from estimating equation 3 clearly shows a strong negative association, indicating that older links are less likely to be terminated. Moreover, the unconditional probability of any link exiting the market is close to 50%, while new links have a conditional probability of exit of nearly 68%. In contrast, 4-year-old links have only a 30% probability of termination, indicating that long-term links tend to be more persistent.

This pattern remains robust across specific years and the cross-section. Furthermore, we also observe this pattern when incorporating link size as a control variable, which indicates that the effect of a link's age extends beyond the survival of larger links. The relationship between link size and exit is presented in Figure 5. This figure shows the creation and destruction rate for each bin in the link size distribution.⁷ The creation rate is defined as the share of links that were inactive the previous year but become active in year t , while the destruction rate is defined as the share of links that are active in year t but will terminate in the following year.

Figure 5: Link Creation and Destruction rates



Note:

Two interesting patterns emerge from Figure 5. First, both creation and destruction rates decrease with link size. Second, only large links exhibit a positive net creation rate. These patterns reinforce previously described facts, depicting an economy with a higher creation of small links that should exit soon, and a lower creation of large links that are more persistent and account for a significant share of the network's value.

⁷I use 20 bins, so that each dot represents 5% of the observations.

II. C Robustness

This section documents a series of robustness exercises designed to verify that the three main empirical facts of the paper—the disproportionate relevance of long-tenured links, the positive tenure-size relationship, and the declining hazard profile—are not artifacts of specific sample definitions, sectoral composition, or unbalanced panels. Each robustness exercise corresponds to a set of figures in the slide deck, and all conclusions below draw directly from those empirical patterns. Across all checks, the core facts remain remarkably stable, supporting the view that link dynamics reflect underlying economic fundamentals rather than mechanical features of the data.

II. C.1 Partial-Year Effects

A first concern is that some relationships may appear short-lived simply because buyers or sellers enter or exit the sample mid-year, creating artificial “partial-year” observations. To address this concern, we recompute all three main facts redefining the age of each link according to the month we observe the first transaction. Therefore, the first year of a buyer-seller link that started trade in March includes all transactions between March of that year and February of the next year.

This restriction ensures that each observed link has the same exposure window for survival, growth, and sales accumulation since birth. It is particularly important for interpreting hazard rates and tenure profiles, which could otherwise be biased if newly formed links are mechanically more likely to appear small or exit quickly because they had less time to trade in the calendar year.

Figures A3–A5 show that the robustness-adjusted patterns closely mirror the baseline results. Long-tenured links continue to account for a disproportionately high share of network sales; the positive tenure-size gradient remains strong; and hazard rates continue to decline sharply with tenure. The similarity of these patterns confirms that partial-year censoring does not drive the main facts.

II. C.2 Balanced Panel of Firms

The baseline sample allows firms to enter and exit the dataset over 2017–2022. Although firm turnover is a natural feature of the economy, unbalancedness could mechanically affect the count of long-tenured links. To ensure that firm dynamics are not confounding link dynamics, we restrict the sample to a *balanced panel* of firms present in all years.

This check is relevant because long-tenured relationships necessarily involve stable firms; thus, the balanced panel isolates whether the main facts arise solely from the

behavior of firms that are always active.

The results in Figures A6–A8 replicate the core patterns. Long-tenured links remain central to total sales, the age-size gradient persists, and survival hazards continue to decline monotonically. This suggests that link dynamics are not an artifact of firm entry or exit but reflect persistent patterns within ongoing relationships.

II. C.3 Exclusion of Retail and Wholesale Sectors

Retail and wholesale firms may engage in large-volume, low-duration transactions that differ structurally from production-oriented relationships. To ensure that these sectors do not disproportionately influence the main facts, we exclude all relationships involving firms classified in retail or wholesale industries.

This robustness check is crucial because these sectors exhibit high buyer-supplier churn by design, and their presence could inflate the turnover rate or alter tenure distributions.

Figures A9–A11 show that the three core empirical facts remain intact. Long-tenured links continue to dominate value shares, the size-tenure profile is largely unchanged, and hazard rates again fall sharply with tenure. This indicates that the patterns arise within the production structure itself and are not driven by retail-wholesale supply chains.

II. C.4 Exclusion of Capital-Intensive Sectors

Because capital-intensive sectors often have longer-lived physical production chains, one may worry that they mechanically generate longer link tenures or larger link sizes. To rule out this explanation, we re-estimate the main facts excluding all capital-intensive industries.

This exercise checks whether the core patterns are driven by sector-specific technological features rather than general link dynamics.

As shown in Figures A12–A14, the main facts survive the exclusion. Long-tenured links remain disproportionately relevant, size rises with tenure, and hazard rates decline. Hence, sector-specific capital intensity cannot account for the empirical patterns.

II. C.5 Cross-Sectional Robustness

Finally, we evaluate whether the main patterns arise only in panel data or whether similar relationships are visible in a single cross section. We re-estimate the tenure-size

and hazard-tenure patterns avoiding the use of the buyer-year and seller-year fixed effects, using cross-sectional variation to estimate the coefficients.

This check is especially important because cross-sectional variation is often used in structural estimation and because tenure proxies may be observed differently in cross sections.

Figures A15–A17 confirm that the key facts emerge even in the cross-sectional data. Link size rises with tenure, and hazard patterns remain monotonic. This demonstrates that the empirical relationships are visible in both longitudinal and static data environments.

II. D Taking Stock

The empirical evidence presented so far show that long-term business links are economically relevant: there are relatively few long-term links, but they contribute disproportionately to the value of the network (Fact 1). The relevance stems from their significantly larger size; long-term links exceed the average new link by more than seven times (Fact 2). A significant portion of this size differential arises from the initial size of the links rather than differences in their growth trajectories. Moreover, long-term links demonstrate higher persistence, exhibiting significantly smaller exit probabilities (Fact 3).

What economic forces account for the relevance and dynamics of long-term links? In the next section, I outline a model that positions selection as the main driver of long-term link relevance. Then, I use the model to quantify the increased value of long-term links.

III Theoretical Framework

This section presents the model used to study the evolution of production networks and its aggregate implications. The model features a static equilibrium with dynamic states, allowing for computational tractability while closely following the evolution of individual business links. For ease of exposure we omit the time subindex from the baseline model.

III. A Environment

The economy is populated by a representative household and a unit continuum of monopolistically competitive firms. Firms are indexed by their productivity type $z \in \mathcal{Z}$,

distributed according to the cumulative distribution function $G(z)$. Each firm produces a differentiated variety that is used both for final consumption and as an intermediate input by other firms.

Households. The representative household has CES preferences over the differentiated varieties supplied by firms:

$$W = \left(\int_{\mathcal{Z}} x(z)^{\frac{\sigma-1}{\sigma}} dG(z) \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1, \quad (4)$$

where $x(z)$ denotes final consumption of the variety produced by a firm of type z , $\mathcal{Z}$ is the set of active firms in the economy, and σ is the elasticity of substitution across varieties. We assume that varieties are imperfect substitutes and therefore $\sigma > 1$.⁸

The household supplies an inelastic aggregate endowment of labor L , owns all firms, and receives their profits. Its budget constraint is

$$\int_{\mathcal{Z}} p(z)x(z) dG(z) \leq wL + \Pi, \quad (5)$$

where $p(z)$ is the price of variety z , w is the wage, and Π denotes aggregate profits.

Firms and Production. Each firm produces output using labor and a composite of intermediate inputs. The production technology of a firm with productivity z is a Cobb-Douglass production function with constant returns to scale

$$y(z) = \kappa z l(z)^{\alpha} v(z)^{1-\alpha}, \quad \alpha \in (0, 1), \quad \kappa > 0, \quad (6)$$

where $l(z)$ denotes labor demand, $v(z)$ is the firm's composite intermediate input, κ is a normalization constant, and z is a Hicks-neutral productivity term.⁹ The parameter α is the labor cost share, while $1 - \alpha$ is the intermediate-input cost share.

To produce, firms combine labor with intermediate inputs purchased from other firms. The intermediate-input composite of a buyer of type z is a CES aggregate over the differentiated varieties supplied by its active trading partners:

$$v(z) = \left[\int_{\mathcal{Z}} \int_{r^*(z, z')}^{\infty} (r v(z, z'; r))^{\frac{\sigma-1}{\sigma}} dH(r) dG(z') \right]^{\frac{\sigma}{\sigma-1}}, \quad (7)$$

where r is the bilateral compatibility or quality of the relationship with stationary distribution H , $v(z, z'; r)$ is the quantity of the input purchased by customer z from supplier

⁸Because preferences are homothetic, expenditure shares depend on relative prices but not on the household's income level.

⁹The parameter κ is given by $\kappa = \alpha^{-\alpha} (1 - \alpha)^{-(1-\alpha)}$ and is used to simplify algebraic expressions in marginal costs.

z' when the underlying link state is r , and $r^*(z, z')$ is the endogenous bilateral compatibility cutoff that determines whether a potential relationship between a buyer of type z and a supplier of type z' is active.

Variable r captures an idiosyncratic, match-specific quality or suitability state that governs the bilateral compatibility, such as technological fit, between a buyer and a potential supplier. A higher match-specific compatibility component $r(z, z')$ makes supplier z' input more effective in the production of buyer z . Thus, relationship heterogeneity affects both the intensive margin (the amount traded conditional on an active relationship) and the extensive margin (whether the relationship is formed). The threshold $r^*(z, z')$ makes the production network endogenous: firms purchase inputs only from the subset of potential suppliers for which link formation is privately profitable, as will be explained in detail below.

The double integral in (7) provides a type-level representation of aggregation across differentiated supplier varieties. The outer integral aggregates across supplier productivity types z' . Conditional on a pair of buyer and supplier types (z, z') , there is an atomless continuum of potential buyer-supplier matches, each characterized by one realized bilateral compatibility r , drawn independently from the distribution H . By the exact law of large numbers, the cross-sectional distribution of realized match qualities within each (z, z') cohort coincides with H . The inner integral aggregates across distinct supplier relationships with different realized values of r . By restricting the integration domain to $r \geq r^*(z, z')$, it includes only profitable relationships.

For tractability, the elasticity of substitution across intermediate inputs is assumed to equal the elasticity of substitution across final-good varieties, σ . This restriction delivers a common demand elasticity in final and intermediate markets and therefore a common monopolistic-competition markup. It can be relaxed by introducing separate elasticities for final consumption and intermediate production.

Market clearing. In this economy, labor can be used for production (L^Y) or to create and maintain the network of firms (L^O). Therefore, the labor market clearing condition is

$$L = L^Y + L^O \quad (8)$$

where $L^Y = \int l(z) dG(z)$ is total labor demanded by firms to produce each differentiated variety, and $L^O = \int \bar{f} l(z, z') dG(z) dG(z')$ is total labor used to maintain and create the production network. Importantly, we assume that the relationship cost is paid only by relationships that are available and active in the current period.

At the same time, a firms output can be consumed by the representative household

$x(z)$, or by other firms as intermediate input $v(z, z')$. Therefore,

$$y(z') = x(z') + \int v(z, z') dG(z) \quad \forall z' \in G(.) \quad (9)$$

III. B Decentralized Equilibrium

We characterize the decentralized equilibrium in two steps. First, we solve for prices, quantities, and input choices conditional on a given production network. We then endogenize the network by allowing firms to choose which buyer-supplier relationships to activate. For simplicity, wage is assumed as the numeraire.

Equilibrium Conditional on the Network. Consider first a fixed production network. At this stage, the cutoff schedule $r^*(z, z')$, and hence the set of active relationships, is taken as given. A relationship operates only if it is selected by the for trade. We characterize its endogenous determination in the next subsection.

Given the wage w (numeraire) and the price charged by suppliers, the marginal cost of a firm of type z is

$$c(z) = \frac{\mathbb{P}(z)^{1-\alpha}}{z} \quad (10)$$

where the ideal price index for the firm's intermediate input composite is given by

$$\mathbb{P}(z) = \left(\int_{\mathcal{Z}} \int_{r^*(z, z')}^{\infty} p(z')^{1-\sigma} r^{\sigma-1} dH(r) dG(z') \right)^{1/(1-\sigma)} \quad (11)$$

Conditional on an active relationship between a buyer of type z and supplier of type z' , expenditure minimization yields the bilateral intermediate input demand

$$v(z, z'; r) = r^{\sigma-1} p(z')^{-\sigma} \mathbb{P}(z)^{\sigma-1} M(z) \quad (12)$$

where $M(z)$ denotes the firms total expenditure on intermediate inputs. A higher bilateral compatibility r both raises the effective quantity delivered per physical unit and, through optimal input substitution, increases the physical quantity purchased from that supplier. Importantly, an unavailable relationship has $v(z, z'; r) = 0$.

Because each firm faces a demand elasticity equal to σ , profit maximization under monopolistic competition implies the constant markup pricing rule

$$p(z) = \frac{\sigma}{\sigma-1} c(z) \quad (13)$$

The equilibrium conditional on the network can be found by computing two fixed points. First, the backward propagation fixed point allow us to solve for input prices

and marginal costs, conditional on the network. Then, the forward propagation fixed point allow us to solve for sales, total expenditures, and bilateral trade, conditional on the network.

The backward propagation equation describes how input prices depends on direct and indirect suppliers in the network, and therefore how shocks propagate downstream from suppliers to buyers. Define the selected quality-adjusted sourcing capacity $q(z, z') = \int_{r^*(z, z')}^{\infty} r^{\sigma-1} dH(r)$ as the expected contribution of active states between the buyer z and the seller z' .¹⁰ Then, we can express the ideal price index of intermediate purchases as

$$\mathbb{P}(z) = \left(\int p(z')^{1-\sigma} q(z, z') dG(z') \right)^{1/(1-\sigma)}$$

Note that by replacing the pricing scheme from equation 13, and the marginal costs from equation 10 into the price index equation 11, we have that input price index for a firm depends on it's suppliers productivity, input price index and sourcing capacity.¹¹

$$\mathbb{P}(z)^{1-\sigma} = \left(\frac{\sigma}{\sigma-1} \right)^{1-\sigma} \int \mathbb{P}(z')^{(1-\alpha)(1-\sigma)} z'^{\sigma-1} q(z, z') dG(z') \quad (14)$$

The forward propagation equation describes how a firm's demand (from the final consumer and other firms as intermediate input) depends on demand from their customers, and therefore how shocks propagate upstream from buyers to suppliers. The forward equation is presented below,¹² where $S(z)$ denote total sales of firm z , P is the final-consumption price index (previously defined), and E is consumer's total expenditure.

$$S(z) = \left(\frac{\sigma}{\sigma-1} \right)^{1-\sigma} z^{\sigma-1} \mathbb{P}(z)^{(1-\sigma)(1-\alpha)} \left(\frac{E}{P^{1-\sigma}} + \frac{1-\alpha}{\sigma/(\sigma-1)} \int S(z') \mathbb{P}(z')^{\sigma-1} q(z', z) dG(z') \right) \quad (15)$$

The remaining variable we need to pin down to solve the equilibrium is nominal income E , which is identified from the labor market clearing and aggregate labor resource constraint. The proof of existence and uniqueness of the equilibrium conditional on the network is in appendix ??.

Before the description of the endogenous network equilibrium, it's important to highlight a few definitions of network variables. In the model, let $\omega(z, z')$ be the CES expenditure share of firm z on the good sold by firm z' , as presented in equation 16.

¹⁰Note that $q(z, z')$ summarizes everything about the distribution of the bilateral compatibility that is relevant for the CES price index, including information about the network of firms. In addition, we assume that $q(z, z') < \infty$, $\forall(z, z')$.

¹¹Equation 14 resembles equation 6 from Bernard et al. (2022).

¹²The details about the derivation of the forward equation can be found in appendix ??.

Let $\tilde{\Omega}$ be the cost-based input-output matrix, where each (z, z') cell measures the direct importance of supplier z' in buyer's z production costs, as presented in equation 17. Let $\tilde{\Psi} = (I - \tilde{\Omega})^{-1}$ be the cost-based Leontief inverse that captures the total direct and indirect cost exposure of firms in the network, and let $\tilde{\Lambda} = b^T \tilde{\Psi}$ be the cost-based Domar weight, where b captures the relevance of each variety for the final consumer.

$$\omega(z, z') = \frac{p(z')^{1-\sigma} q(z, z')}{\mathbb{P}(z)^{1-\sigma}} \quad (16)$$

$$\tilde{\Omega}(z, z') = \frac{\partial \log c(z)}{\partial \log p(z')} = (1 - \alpha) \omega(z, z') \quad (17)$$

Endogenous Network Equilibrium. Consider now the case of firms optimally selecting their trade partners. We assume that sellers must pay a fixed maintenance cost $\bar{f}$ to trade with each active buyer they have. Together with the CES structure of the model and assuming that goods are substitutes (i.e., $\sigma > 1$), sellers treat each customer independently, which helps with the tractability of the network.¹³

The seller decision to start or maintain a relationship with a buyer will depend on that specific link profitability and the fixed maintenance cost $\bar{f}$.¹⁴ Thus, sellers will only keep profitable relationships. Let $\pi(z, z', r)$ be the variable profits if firm z' sells to firm z when the match-state is r . Given the assumptions of the model, we can express the variable profits of an individual link as a σ -fraction of the bilateral sales of the link $m(z, z', r) = p(z')v(z, z', r)$. Therefore, the share of firms z' that sell their good to firms z is given by

$$\ell(z, z') = \int \mathbb{1} \left\{ \frac{m(z, z', r)}{\sigma} \geq \bar{f} \right\} dH(r) \quad (18)$$

Note that equation 18 implies that for each bilateral pair, there is a compatibility threshold $r^*(z, z')$, or the minimum underlying link state that makes the bilateral relationship active. Moreover, given the distribution of r , we can find the share of active trading pairs as follows

$$\ell(z, z') = 1 - H(r^*(z, z')) \quad (19)$$

We can identify the threshold $r^*(z, z')$ by replacing the expression of bilateral sales into equation 18. For any bilateral trade, the cutoff is given by

$$r^*(z, z')^{\sigma-1} = \underbrace{\sigma}_{\text{Inverse Private Profit Share}} \underbrace{\left(\frac{q(z, z')}{(1 - \alpha) \omega(z, z')} \right)}_{\text{Network Composition}} \underbrace{\left(\frac{\bar{f}}{\tilde{\Lambda}(z) L^Y} \right)}_{\text{Link cost relative to Buyer Scale}} \quad (20)$$

¹³If buyers pay a fixed maintenance cost, even if firms are gross substitutes we would have complementarities in the decision to create links because links directly affect marginal costs.

¹⁴Note that because of love-of-variety effects, buyers will always prefer to have more suppliers.

Equation (20) characterizes the bilateral compatibility threshold for an active relationship between any pair of firms. A potential link between a buyer of type z and a supplier of type z' is formed if and only if its compatibility satisfies $r \geq r^*(z, z')$. The cutoff equates the operating profit generated by the marginal relationship to the fixed labor cost required to maintain it.

The first component captures private appropriability. Under CES demand and monopolistic competition, the supplier appropriates only the fraction $1/\sigma$ of bilateral revenue as operating profit. Consequently, a higher elasticity of substitution reduces the private profit generated by a given relationship and raises the productivity required for the link to be viable.

The second component captures the composition of the buyer's sourcing network. The variable $q(z, z')$ measures the quality-adjusted mass of active relationships between types (z, z') , while $\omega(z, z')$ is the corresponding input-expenditure share. Their ratio maps the productivity of an individual relationship into its contribution to the buyer's total intermediate expenditure. Holding $q(z, z')$ fixed, a larger expenditure share $\omega(z, z')$ makes a relationship with suppliers of type z' more valuable and lowers the cutoff. Conversely, holding the expenditure share fixed, a larger quality-adjusted mass of competing relationships dilutes the contribution of any individual link and raises the cutoff.

The final component compares the fixed cost of maintaining a relationship with the economic scale of the buyer. The cost-based network weight $\tilde{\Lambda}(z)$ captures the demand reaching a type- z firm through both final and intermediate uses, while L^Y determines the aggregate scale of variable production. Relationships generate more bilateral revenue when the buyer is larger or more central in the production network. Thus, a larger $\tilde{\Lambda}(z)L^Y$ lowers the cutoff and supports a denser supplier network, whereas a higher fixed cost $\bar{f}$ raises the cutoff and reduces link formation.

The equilibrium network is characterized as a fixed point in the cutoff schedule. Given a conjectured quality-adjusted network $q(z, z')$, the backward and forward equilibrium systems determine prices, sales, expenditure shares, cost-based network weights, and aggregate production scale. These objects imply an updated cutoff schedule through equation (20). The updated cutoffs, in turn, determine both the share of active relationships,

$$\ell(z, z') = 1 - H(r^*(z, z'))$$

and the quality-adjusted sourcing capacity,

$$q(z, z') = \int_{r^*(z, z')}^{\infty} r^{\sigma-1} dH(r)$$

An endogenous-network equilibrium is a fixed point at which the cutoff schedule,

and hence the induced network objects, remain unchanged.

IV Model Estimation

We estimate the model to quantify the contribution of bilateral compatibility to relationship dynamics and to evaluate the aggregate consequences of endogenous network selection. The parameter vector is

$$\Theta = \left\{ \sigma, \alpha, L, \bar{f}, \rho_z, \sigma_z, \rho_r, \sigma_r \right\}, \quad (21)$$

where σ is the elasticity of substitution, α is the labor share in variable production costs, L is aggregate labor supply, and $\bar{f}$ is the labor cost of maintaining a buyer–supplier relationship. The remaining parameters characterize the stochastic processes for firm productivity, z , and bilateral compatibility, r .

We combine external calibration with simulated method of moments. Three parameters are calibrated outside the model: the elasticity of substitution σ , the labor cost share α , and aggregate labor supply L .

In the baseline specification, we set $\sigma = 5$ and examine robustness to alternative values. This value lies within the range commonly used in quantitative production-network models. Under CES demand and monopolistic competition, σ determines both the elasticity of firm demand and the equilibrium markup.¹⁵

We set $\alpha = 0.24$ to match the aggregate labor share in gross output, with the remaining share, $1 - \alpha$, corresponding to intermediate-input expenditures. A lower value of α increases the importance of intermediate inputs in production and therefore strengthens the aggregate effects of supplier quality and production-network adjustment. Finally, we normalize aggregate labor supply to $L = 100$. This normalization fixes the scale of the economy without affecting relative prices or expenditure shares.

The estimation of the remaining parameters is explained below.

IV. A Structural Estimation

We estimate the parameters governing firm productivity using a regression, and the parameters governing the bilateral compatibility and relationship costs by simulated method of moments. The targeted moments summarize the cross-sectional and dynamic properties of buyer-supplier relationships.

¹⁵Thus, $\sigma = 5$ implies a markup of 1.25 over marginal cost and an operating-profit share of revenue equal to $1/\sigma = 0.20$, before accounting for relationship costs.

We assume that firms' productivity follows a stationary AR(1) process:

$$\log z_{i,t} = \rho_z \log z_{i,t-1} + \varepsilon_{i,t}^z, \quad \varepsilon_{i,t}^z \sim \mathcal{N}(0, \sigma_z^2) \quad (22)$$

where ρ_z determines the persistence of firm productivity and σ_z determines the dispersion of its innovations. The stationary distribution induced by equation (22) corresponds to $G(z)$ in the theoretical model.

We estimate the firm-productivity process outside the simulated method of moments. Specifically, we estimate a regression of labor productivity on its lag, where we set ρ_z equal to the estimated coefficient, and σ_z equal to the standard deviation of the regression residual. Standard errors are clustered at the firm level to allow for serial correlation and heteroskedasticity within firms. This approach provides a transparent empirical discipline for the productivity process and reduces the dimensionality of the structural estimation. Its main limitation, however, is that labor productivity is not a direct measure of Hicks-neutral productivity: it also reflects input intensity, prices, demand, and endogenous supplier quality. Moreover, measurement error may attenuate persistence, while unobserved permanent firm heterogeneity may increase it.

We assume that the bilateral compatibility evolves according to

$$\log r_{ij,t} = \rho_r \log r_{ij,t-1} + \varepsilon_{ij,t}^r, \quad \varepsilon_{ij,t}^r \sim \mathcal{N}(0, \sigma_r^2) \quad (23)$$

where ρ_r governs the persistence of relationship fundamentals and σ_r governs the dispersion of the match-specific innovations. Its stationary distribution corresponds to $H(r)$. Innovations to firm productivity and bilateral compatibility are assumed to be independent across firms and relationships and mutually independent. This assumption isolates relationship-specific variation from persistent changes in buyer or supplier productivity.

We identify the parameters of the match-specific compatibility shifters using moments from bilateral expenditures. Conditional on an active relationship, firm-to-firm sales satisfy

$$m_{ij,t} = r_{ij,t}^{\sigma-1} p_{j,t}^{1-\sigma} \mathbb{P}_{i,t}^{\sigma-1} M_{i,t}$$

Taking logarithms gives

$$\log m_{ij,t} = \underbrace{\log M_{i,t} + (\sigma - 1) \log \mathbb{P}_{i,t}}_{\text{buyer-year component}} + \underbrace{(1 - \sigma) \log p_{j,t}}_{\text{supplier-year component}} + (\sigma - 1) \log r_{ij,t}.$$

This expression motivates a buyer-year and supplier-year fixed-effects decompo-

sition

$$\log m_{ij,t} = \gamma_{i,t}^B + \gamma_{j,t}^S + u_{ij,t}, \quad (24)$$

where, according to the model,

$$u_{ij,t} = (\sigma - 1) \log r_{ij,t}. \quad (25)$$

Buyer-year fixed effects absorb changes in the buyer's intermediate expenditure and input price index, while supplier-year fixed effects absorb changes in the supplier's price.

Equation (24) is observed only for active relationships. Consequently, the empirical distribution of $u_{ij,t}$ is selected by the endogenous cutoff $r_{ij,t} \geq r_{ij,t}^*$. Rather than treating the fixed-effects residuals as an unselected sample from H , the estimation procedure reproduces this selection within the simulated model. This is important because relationships with low bilateral compatibility are systematically less likely to be observed and have shorter durations.

It is important to note the similarities between equation (24) and equation (1), the regression we use to describe the empirical patterns. The structural decomposition provides the formal economic mechanism behind the empirical patterns documented in equation (1). By projecting the bilateral expenditure residuals from equation (24) onto link age, we construct age-size gradients that isolate the evolution of the latent match state. According to the structural model, the positive association between link size and age observed in the transaction data is driven exclusively by the dynamic selection of the bilateral compatibility term, $r_{ij,t}$. Consequently, the empirical age-size gradients generated by equation (1) serve as ideal targeted moments in the SMM procedure, tightly identifying the persistence ρ_r and innovation dispersion σ_r of the underlying process.

Similarly, to construct the empirical hazard rate moments, we estimate the probability of relationship destruction conditional on link tenure. Through the lens of the structural model, a link is severed when the latent bilateral compatibility $r_{ij,t}$ falls below the endogenous survival threshold $r^*(z_{i,t}, z_{j,t})$. Because this compatibility fundamental follows a persistent AR(1) process, relationships that survive for multiple periods are positively selected. This selection implies that older links face a systematically lower probability of receiving a sequence of negative shocks large enough to trigger relationship destruction. Therefore, matching the empirically observed decreasing age-hazard profile in the SMM procedure provides another source of variation for the identification of the persistence ρ_r and volatility σ_r parameters of the bilateral compatibility process.

Table 1 shows a summary of each parameter and the moment that helps with its identification. The moments play complementary identification roles. Average creation rate primarily disciplines $\bar{f}$ relative to the location of the bilateral compatibility distri-

bution. The bilateral expenditure-age profile disciplines ρ_r , while the frequency with which relationships cross the endogenous cutoff discipline σ_r .

Table 1: Parameter Identification

Parameter	Identifying moment or source	Identification logic
ρ_r	Link size-age profile	Persistence of bilateral compatibility
σ_r	Hazard-age link profile	Selection
$\bar{f}$	Average link probability	Average creation rate

Notes: The table reports the primary source of identification for each parameter. Identification is joint because firm sales, bilateral expenditures, and network formation are simultaneously determined in equilibrium. Bilateral expenditure residuals are obtained after absorbing buyer-year and supplier-year fixed effects.

Let $\mathcal{M}^D$ denote the vector of empirical moments and $\mathcal{M}^S(\Theta)$ the corresponding moments generated by the model. The estimator solves

$$\hat{\Theta} = \arg \min_{\Theta} \left[\mathcal{M}^D - \mathcal{M}^S(\Theta) \right]' \mathcal{W} \left[\mathcal{M}^D - \mathcal{M}^S(\Theta) \right], \quad (26)$$

where $\mathcal{W}$ is a positive-definite weighting matrix. In the baseline specification, we use a diagonal matrix based on the inverse of the square of each moment, so that they are scaled.

For a candidate parameter vector, the model is simulated as follows. First, we discretize the stationary firm-productivity distribution G and approximate the bilateral compatibility distribution H using numerical quadrature. Second, given a conjectured cutoff schedule, we solve the backward fixed point for prices and marginal costs and the forward fixed point for sales and intermediate expenditures. Third, the resulting equilibrium objects determine an updated cutoff schedule. We iterate over these steps until the cutoff schedule and quality-adjusted sourcing capacities converge.

To construct the link-age moments, we simulate the processes in equations (22) and (23) for a large panel of potential buyer-supplier matches. A relationship is active whenever

$$r_{ij,t} \geq r^*(z_{i,t}, z_{j,t}). \quad (27)$$

The simulated panel is initialized from the stationary distributions and run for a burn-in period before calculating the model moments. This ensures that creation, destruction, and tenure moments are computed from the stationary distribution of firms and relationships rather than from arbitrary initial conditions.

IV. B Model Fit and Validation

We evaluate model fit using both targeted and untargeted moments. Targeted moments establish whether the model reproduces turnover and bilateral expenditure dynamics. Untargeted moments assess whether the estimated bilateral compatibility process reproduces richer features of the data, including the assortative matching by firm size, and the concentration of purchases across suppliers.

As an additional mechanism test, we compare the baseline model with an alternative specification in which relationship dynamics are generated by shocks to fixed relationship costs rather than bilateral compatibility. The comparison evaluates whether the compatibility model more accurately reproduces the observed monotonic relationship between bilateral expenditure, tenure, and subsequent link destruction.

Estimated Parameters

Table 2 reports the point estimates for the parameters governing the dynamic structural processes of the model. The estimated process for firm productivity has a persistence parameter of $\rho_z = 0.87$ and an innovation standard deviation of $\sigma_z = 0.57$. These values are tightly aligned with the standard firm dynamics literature, which typically finds annual persistence estimates between 0.70 and 0.85.¹⁶

Crucially, the estimated process for bilateral compatibility reveals that relationships are highly structural rather than transient. We estimate a persistence of $\rho_r = 0.88$ and an innovation dispersion of $\sigma_r = 1.22$. The relatively high volatility of the match-specific innovations ($\sigma_r = 1.22$) helps the model to generate the massive gross churn and high initial hazard rates observed in the transaction data.

Finally, the fixed relationship maintenance cost is estimated to be $\bar{f} = 11.82$. Hence, in equilibrium, the estimated parameters imply that nearly $x\%$ of the total labor endowment is used for relationship costs.

Figure 6 presents the targeted empirical moments (red squares) alongside the simulated moments from the estimated model (blue circles). Overall, the model successfully captures the core features of buyer-supplier relationships, tightly replicating both the intensive-margin growth and extensive-margin selection patterns observed in the data.

The right panel demonstrates that the model tracks the empirical residual size gradient almost perfectly. Conditional on survival, both the simulated and empirical relationships grow monotonically, expanding by approximately 1.5 log points over their

¹⁶For example, De Loecker and Syverson (2021) note that typical persistence estimates for firm productivity fall within this range using annual data.

Table 2: Estimated Parameters for Productivity Processes and Network Costs

Parameter	Description	Estimate
<i>Firm Productivity Process</i>		
ρ_z	Persistence of firm productivity	0.87
σ_z	Dispersion of firm productivity innovations	0.57
<i>Bilateral Compatibility Process</i>		
ρ_r	Persistence of bilateral compatibility	0.88
σ_r	Dispersion of bilateral compatibility innovations	1.22
<i>Network Maintenance</i>		
$\bar{f}$	Fixed relationship maintenance cost	11.82

Notes: This table reports the point estimates for the structural parameters governing firm-level productivity and relationship dynamics. The firm productivity (z) and bilateral compatibility (r) states follow independent AR(1) processes in logs, governed by their respective persistence (ρ) and innovation dispersion (σ) parameters. The location parameter for bilateral compatibility (μ_r) is normalized to zero to identify the fixed relationship maintenance cost $\bar{f}$, which is expressed in units of labor.

first five years relative to their initial size. This precise alignment confirms that the estimated AR(1) process for bilateral compatibility accurately translates into the observed expenditure growth of maturing links.

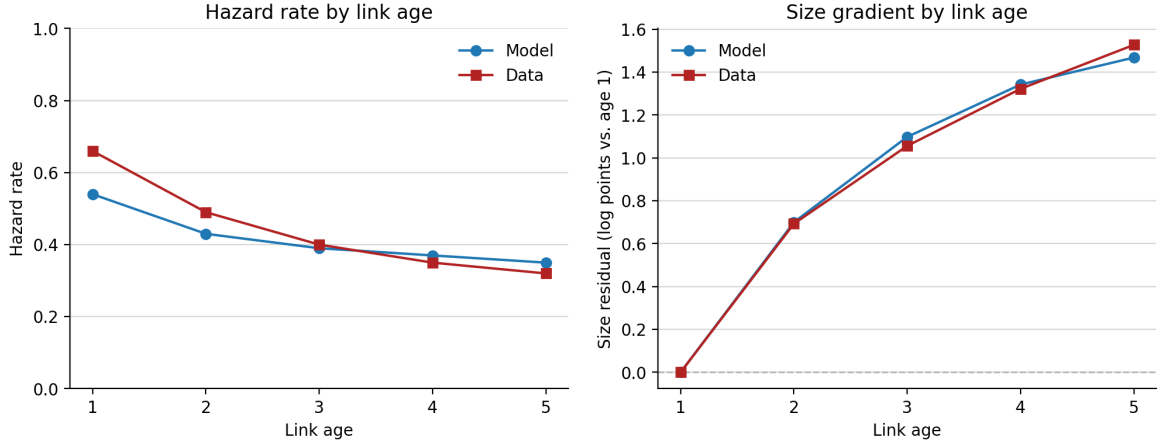
The left panel shows that the model successfully generates the steeply declining hazard rate by link age, which is the central signature of our dynamic selection mechanism. As relationships mature, the surviving links are positively selected, they represent draws situated securely above the continuation threshold, making them significantly less likely to be severed by subsequent negative shocks. Although the model slightly underpredicts the initial destruction rate at age 1, and overpredicts the survival trajectory for established relationships.

Untargeted Model Fit

To further validate the model's structural mechanisms, we evaluate its performance on a set of untargeted moments characterizing the distribution of firm size and network connectivity. Because the theoretical economy is simulated over a unit continuum of firms and scaled to a normalized aggregate labor supply, we standardize both the empirical and simulated variables before comparison.

We first examine the distribution of firm sales within the production network. Although the parameters of the fundamental firm productivity process (z) were estimated independently via an autoregressive panel regression, the cross-sectional distribution of equilibrium sales depends on the full general equilibrium propagation through the endogenous input-output structure. The simulated economy successfully

Figure 6: Model Fit



Notes: This figure evaluates the fit of the Simulated Method of Moments (SMM) estimation by comparing targeted empirical moments (red squares) against the corresponding moments generated by the simulated model (blue circles). The left panel plots the hazard rate as a function of link age. The right panel plots the residual size gradient, defined as the average log-point difference in bilateral transaction volume relative to a relationship at age one once we control for buyer and seller unobserved characteristics. The alignment between the model and the data demonstrates that the estimated bilateral compatibility process successfully replicates both the declining exit probability and the monotonic size growth characteristic of surviving relationships.

replicates the highly skewed, heavy-tailed distribution of standardized firm sales observed in the transaction data. This confirms that propagating exogenous productivity draws through our CES aggregation and endogenous extensive margin framework naturally reproduces the empirical firm size distribution without requiring ad-hoc scaling factors.

Next, we evaluate the unweighted topology of the network by comparing the distribution of the number of customers (out-degree) and the number of suppliers (in-degree) across firms. The simulated economy accurately captures the extreme heterogeneity in network connectivity, which is characterized empirically by a massive density of firms with very few trading partners and a thin, highly dispersed right tail of highly connected hubs. By successfully reproducing these untargeted degree distributions, the model demonstrates that the interaction between fundamental firm productivity and the dynamic bilateral compatibility thresholds naturally generates the hub-and-spoke network topology observed in the Chilean data.

Finally, we provide a stringent test of the model's endogenous selection mechanics by examining the joint distribution of firm size and network connectivity, specifically targeting the standardized number of customers and suppliers across deciles of firm sales. The empirical data exhibits a strong positive gradient, where the largest firms in the top deciles systematically maintain exponentially more relationships than firms in the bottom deciles. The model reproduces this steep, monotonic relationship precisely; because high- z firms possess lower marginal costs and higher intermediate

input demand, they endogenously clear the bilateral compatibility threshold r^* with a much wider measure of potential partners. This close fit validates the model's core theoretical claim: a firm's extensive margin network expansion is inherently tied to its fundamental operating scale.

V Welfare and the Propagation of Shocks

Having characterized the decentralized equilibrium, we now examine how firm-level shocks propagate through the production network and affect aggregate welfare. Because household preferences are homothetic and labor supply is inelastic, welfare is equal to real aggregate expenditure:¹⁷

$$W = \frac{E}{P} \quad (28)$$

where E denotes nominal household expenditure and P is the final-consumption price index.

Endogenous network adjustment introduces a resource tradeoff. Activating additional relationships expands firms' access to productive inputs, but also requires labor for relationship maintenance, reducing the labor available for variable production. Proposition 1 characterizes the first-order welfare effect of firm-level productivity shocks.

Proposition 1: *With wages as the numeraire, changes in welfare are given by*

$$d \log W = d \log L^Y + \tilde{\Lambda}^T \left[d \log z + \frac{1 - \alpha}{\sigma - 1} d \log Q \right]$$

where $d \log Q$ is the change in quality-adjusted sourcing capacity.

This result summarizes the key channels through which endogenous network selection affect welfare in the economy. When the network is fixed, only the second channel is present. Then, the effect on welfare can be summarized by the cost-based Domar weight, and there are no changes in the macro-allocation of labor, as in [Baqae and Farhi \(2019\)](#).

When the network adjusts endogenously, productivity shocks also change relationship cutoffs. The resulting creation and destruction of links affect welfare through both the resources used in network maintenance, and firms' quality-adjusted access to suppliers. The resource channel of welfare changes is captured by the first term, $d \log L^Y$.

¹⁷Since final consumption is the only final use in this closed economy, welfare also coincides with the (model-consistent) real GDP.

The quality-adjusted network change is captured by the last term, $d \log Q$.

The effect of the quality-adjusted network change is amplified by technology assumptions and the network. First, the quality-adjusted change is propagated directly and indirectly through the supply chain, which is captured by the cost-based Domar weight $\tilde{\Lambda}$. Second, the change in sourcing capacity is influenced by the share that firms spend in intermediate inputs $(1 - \alpha)$, and the marginal increase it generates in marginal costs $1/(\sigma - 1)$.¹⁸

Finally, the change in the quality-adjusted sourcing capacity is presented in equation 29. The firm-level change in the sourcing capacity depends on how relevant each supplier z' is for the production process of buyer z , which is captured by the CES shares $\omega(z, z')$. The term $d \log q(z, z')$ measures the change in the bilateral sourcing capacity, and is decomposed next.

$$d \log Q(z) = \int \omega(z, z') d \log q(z, z') dG(z') \quad (29)$$

Lemma 1: *Define the share of active relationships between buyers of type z and suppliers of type z' as equation 19 and the average productivity contribution of an active relationship as*

$$\bar{r}(z, z') \equiv \mathbb{E} \left[r^{\sigma-1} \mid r \geq r^*(z, z') \right]. \quad (30)$$

Then quality-adjusted sourcing capacity satisfies

$$q(z, z') = \ell(z, z') \bar{r}(z, z'), \quad (31)$$

and its change admits the exact decomposition

$$d \log q(z, z') = \underbrace{d \log \ell(z, z')}_{\text{link expansion}} + \underbrace{d \log \bar{r}(z, z')}_{\text{match-quality composition}}. \quad (32)$$

Lemma 1 separates changes in sourcing capacity into an extensive margin and a composition margin. The first term captures changes in the mass of active relationships. The second captures changes in the average productivity of those relationships. When the cutoff falls, more relationships become active, so $d \log \ell(z, z') > 0$. However, the newly activated relationships have lower match productivity than existing relationships, reducing average active-link quality and implying $d \log \bar{r}(z, z') < 0$. The overall effect on $q(z, z')$ is nevertheless positive because each additional relationship makes a positive contribution to sourcing capacity.

Corollary 1: The Composition Response. *Holding the distribution H fixed, suppose*

¹⁸Note that the increase in marginal cost takes the form of love-of-variety effect on sourcing capacity.

that H admits a continuous density h and that the cutoff is positive, with $\ell(z, z') > 0$ and $0 < q(z, z') < \infty$. Define the ratio of the marginal relationship's productivity contribution to the average contribution of active relationships as

$$\theta(z, z') \equiv \frac{r^*(z, z')^{\sigma-1}}{\bar{r}(z, z')}. \quad (33)$$

Changes induced by a change in the cutoff satisfy

$$d \log q(z, z') = \theta(z, z') d \log \ell(z, z'), \quad (34)$$

$$d \log \bar{r}(z, z') = -[1 - \theta(z, z')] d \log \ell(z, z'). \quad (35)$$

Since active relationships satisfy $r \geq r^*(z, z')$, $0 < \theta(z, z') \leq 1$.

Proof. Suppressing the buyer–supplier indices, differentiation gives $d\ell = -h(r^*) dr^*$ and $dq = -(r^*)^{\sigma-1} h(r^*) dr^*$. Using $q = \ell \bar{r}$ yields $d \log q = [(r^*)^{\sigma-1} / \bar{r}] d \log \ell$. The composition response follows from $d \log \bar{r} = d \log q - d \log \ell$. $\square$

The corollary quantifies the composition margin identified in Lemma 1. When firms expand their supplier base by lowering the cutoff, the additional relationships contribute less than the average existing relationship. Consequently, a proportional increase in link mass produces a smaller proportional increase in sourcing capacity. The ratio $\theta(z, z')$ measures the fraction transmitted into sourcing capacity, while $1 - \theta(z, z')$ measures the offset due to declining average match quality. Conversely, when the cutoff rises, the increase in average quality partly offsets the loss of relationships.

V. A Quantifying Endogenous Network Adjustment

We quantify the aggregate relevance of endogenous network adjustment using a productivity counterfactual around the calibrated stationary equilibrium. Holding the population of firms and the distribution of relationship fundamentals fixed, we simulate a large, targeted negative aggregate shock by reducing the fundamental productivity z of the top 5% of firms by 20%.

We perform two exercises to quantify the relevance of network adjustment. In the first exercise, we solve for the welfare impact in steady state. In this case, we assume a continuum of firms and shock only a measure of those firms. The resulting counterfactual is weighted by the baseline measure $dG(z)$, implying the economy goes back to the same steady state distribution. In the second exercise, we analyze how the shock affect the transition path to the steady state. In this case, we simulate an economy with a large but finite number of firms, and compute welfare before and after the productivity shock.

V. A.1 Steady state impact

We evaluate the role of endogenous network formation by comparing the economy before and after a negative firm-productivity shock. We consider two post-shock equilibria. In the fixed-network counterfactual, the set of active relationships remains at its pre-shock configuration, while prices, sales, and aggregate expenditure adjust to the new productivity levels. In the endogenous-network counterfactual, firms can additionally reselect their trading relationships. Both counterfactuals therefore incorporate general-equilibrium adjustments; their difference isolates the additional welfare effect of network reselection. The exercise compares stationary allocations rather than tracing the transition between them.

Let W_0 denote baseline welfare and let W_1^F and W_1^E denote counterfactual welfare under fixed and endogenous cutoffs, respectively. The aggregate contribution of endogenous network adjustment is

$$\mathcal{N} \equiv \log \left(\frac{W_1^E}{W_0} \right) - \log \left(\frac{W_1^F}{W_0} \right) \quad (36)$$

A positive value indicates that adjustment mitigates the welfare loss, while a negative value indicates amplification. Welfare in each equilibrium is computed directly as $W = E/P$. We express welfare changes in consumption-equivalent percentage terms as $100[\exp(\Delta \log U) - 1]$.

We simulate a large, targeted negative aggregate shock by reducing the fundamental productivity z of the top 5% of firms by 20%. Results are presented in table 3. Relative to the pre-shock baseline, welfare falls by 42.13% when the network is fixed and by 42.11% when relationships adjust endogenously. Thus, network reselection slightly mitigates the adverse productivity shock. The endogenous-network allocation delivers 0.04% higher real consumption than the fixed-network allocation. Expressed relative to baseline welfare, this corresponds to a reduction of approximately 0.02 percentage points in the welfare loss.

The welfare improvement accompanies an expansion in relationship activity. The mean link probability across grid type pairs increases from 0.054 to 0.056, while maintenance labor rises by approximately 17% relative to the fixed-network counterfactual. The welfare comparison accounts for both the productive consequences of reselection and the additional resources devoted to maintaining relationships. Under this calibration and shock, the net benefit of endogenous adjustment is positive but small relative to the overall welfare loss.

Table 3: Welfare Comparison: Top 5% Firm Productivity Shock

Scenario	log Welfare	CE Welfare (%)	Network Density	log L_{maint}
Baseline	24.33	0.0000	0.054	-1.39
Fixed network	23.78	-42.13	0.054	-1.39
Endogenous network	23.78	-42.11	0.056	-1.23
Network margin	–	0.04	0.002	0.16

Notes: This table presents the steady-state general equilibrium responses to a 20% productivity reduction among the top 5% of firms. The CE Welfare column reports the consumption-equivalent welfare change relative to the baseline. The network margin isolates the macroeconomic buffering effect of the extensive margin by differencing the endogenous and fixed network outcomes.

V. A.2 Transition Path

We complement the stationary counterfactual with a transitory productivity-shock exercise. We simulate 2,000 firm-productivity histories, initialized from the stationary distribution, and solve the equilibrium over ten pre-shock periods and fifteen post-shock periods. At date $t = 0$, we reduce log productivity by 0.2 for firms in the top 5% of the productivity distribution at date $t = -1$. Treated identities remain fixed thereafter, and the intervention decays according to the productivity process. Prices, sales, and labor allocation are re-solved each period. Match quality follows its calibrated process and is integrated analytically, rather than represented by individual bilateral draws.

We construct four paths, combining endogenous or fixed relationship selection with the presence or absence of the productivity intervention. The paths share the same underlying productivity innovations. In the fixed-network scenarios, relationship selection is frozen at date $t = -1$, although the productive contribution of inherited relationships evolves with match quality. In the endogenous-network scenarios, firms update their relationship choices each period. For each regime, the shock response subtracts its own no-shock control, thereby separating the intervention from the ongoing consequences of productivity and match-quality evolution. We report pointwise consumption-equivalent responses as $100[\exp(\log U_t^{\text{shock}} - \log U_t^{\text{control}}) - 1]$, rather than discounted lifetime welfare changes.

The productivity intervention generates a large initial welfare gain that subsequently declines. On impact, real consumption increases by 91.78% under endogenous selection and 91.98% under fixed selection, each relative to its corresponding control. By period $t = 5$, these responses decline to 8.77% and 14.75%, respectively; at the final reported date, $t = 14$, they are 0.73% and 4.07%. The responses therefore substantially dissipate over the reporting horizon, although neither the intervention nor its welfare effects have completely vanished.

Endogenous selection reduces the proportional welfare response to this positive

intervention relative to the fixed-network regime. The difference between the two log responses, converted to consumption-equivalent units, is approximately -0.11% on impact, reaches -5.29% at $t = 12$, and equals -3.21% at $t = 14$. These differences compare shock responses relative to distinct control paths; they do not imply that the endogenous-network allocation delivers lower welfare levels. Indeed, allowing continued reselection can improve the no-shock allocation as inherited relationships become less well suited to evolving economic conditions. The exercise therefore distinguishes the welfare value of ongoing reselection from its incremental contribution to the response to a particular productivity intervention.

VI Efficiency of Endogenous Network Formation

Having analyzed how endogenous network formation shapes the network and propagation of shocks, we now turn to understand efficiency. Consider the case of a social planner deciding the production network and the allocation of resources in the economy. The planner has two distortions to consider. First, firms charge a constant markup over marginal costs that distort pricing and quantity decisions. This distortion adds up since we have a production network, which further distort firms' decision from social optimum given the double marginalization problem. And second, firms decision to

The decentralized equilibrium of this economy is inefficient due to the interaction between monopolistic competition and the fixed resource costs of relationship maintenance. These frictions introduce welfare-reducing wedges along two distinct margins: the intensive margin of intermediate trade and the extensive margin of endogenous network formation. Along the intensive margin, standard constant markup pricing under CES aggregation drives a continuous wedge between the social marginal cost of production and the private prices faced by downstream buyers. Because suppliers extract a positive markup on every unit sold, downstream firms face artificially high marginal costs, resulting in sub-optimally low transaction volumes conditional on a relationship being active.

The more complex distortion operates on the extensive margin of network formation. When a buyer and supplier evaluate a potential relationship, the link is formed only if the private bilateral operating profits strictly exceed the fixed maintenance cost $\bar{f}$. However, because the buyer evaluates this link subject to the supplier's markup, the buyer's private reduction in marginal cost fails to fully internalize the total social surplus generated by the match. Specifically, the linking pair ignores the downstream "love-of-variety" benefits and the general equilibrium profit expansion that their new connection generates for the rest of the supply chain. Consequently, the decentralized profitability threshold $r^*(z, z')$ is systematically higher than the social planner's optimal

compatibility cutoff, meaning the decentralized production network is sub-optimally sparse.

Evaluating this extensive margin inefficiency is generally intractable in arbitrary networks due to the highly non-linear, pair-specific propagation of local network centralities. However, under the structural assumption that bilateral compatibility r is Pareto-distributed, the planner's problem simplifies dramatically. The Pareto parameterization guarantees a "global sparsity" result where the complex, endogenous network centrality responses mathematically cancel out. This property elegantly collapses the pair-specific extensive margin distortions into a single, closed-form global inefficiency wedge. Ultimately, this demonstrates that the aggregate efficiency loss from decentralized network formation is determined entirely by the macroeconomic network sensitivity parameter η relative to the markup, allowing the social planner's optimal allocation to be characterized without computing the full matrix of heterogeneous network linkages.

VII Conclusion

This paper provides new evidence and a theoretical framework to understand the microeconomic forces shaping the evolution of production networks. Using comprehensive firm-to-firm transaction data from Chile, the empirical analysis demonstrates that long-term business relationships, though relatively rare, account for the majority of economic activity in the network. These persistent relationships are larger, grow over time, and exhibit substantially lower exit rates, highlighting the importance of dynamic selection in shaping the observable distribution of link sizes and lifespans. The empirical facts are robust across multiple dimensions of the data, confirming that they are not artifacts of sectoral composition, unbalanced panels, or partial-year censoring. Instead, they reveal fundamental properties of firm-to-firm relationships in modern production systems.

Motivated by these patterns, the paper develops a tractable general equilibrium framework in which link continuation is governed by either a dynamic fixed-cost process or a dynamic match-specific productivity shifter. Embedding these microfoundations into a CES production structure allows for a clear characterization of how firm-to-firm trade flows, marginal costs, and input price indices respond to match-specific heterogeneity. The analysis shows that the two sources of dynamics lead to fundamentally different selection forces. In the Fixed Cost model, link survival depends on transitory cost shocks unrelated to economic fundamentals, generating a network characterized by high turnover and little within-link evolution. In contrast, the Productivity Shifter model produces smooth and persistent heterogeneity, whereby link size

evolves alongside match quality and selection gradually eliminates relationships with low quality-adjusted prices.

The Monte Carlo simulations reveal that these differing mechanisms produce sharply different conditional moments. The Fixed Cost model generates flat or declining link tenure profiles, inconsistent with the empirical evidence, because equilibrium trade flows do not respond to cost shocks. The Productivity Shifter model, by contrast, generates a strong positive tenure-size gradient and persistent heterogeneity, aligning closely with the data. These results underscore the importance of modeling link-specific productivity dynamics to replicate the empirical regularities in production networks. They also show that different microfoundations of link formation imply different forms of misallocation, different speeds of adjustment, and ultimately different aggregate TFP paths.

The aggregate implications of the two models differ systematically. The Fixed Cost model implies excessive link churn, the destruction of productive relationships, and heightened volatility, making it an upper bound for misallocation induced by link dynamics. The Productivity Shifter model, on the other hand, produces more efficient selection, smoother adjustment, and a richer interplay of extensive and intensive margins, suggesting lower welfare losses from network restructuring. These contrasts emphasize that the modeling choice governing link evolution is central for quantification: distinct microfoundations imply different propagation of shocks through supply chains and different aggregate outcomes.

Taken together, the empirical evidence, theoretical framework, and simulation results highlight the need for structural models that incorporate persistent match-specific heterogeneity and endogenous selection to account for the dynamics of production networks. Future work will extend the analysis in several directions. A central next step is to estimate the structural parameters governing both firm and match-level AR(1) processes using the Chilean data, allowing for disciplined quantification of the relative importance of fixed costs versus productivity-based selection. Another priority is to compute welfare and TFP paths under the two microfoundations to measure the aggregate efficiency consequences of link churn. A deeper analysis of the planner's problem will also help assess whether decentralized link formation leads to excessive turnover or insufficient experimentation. More broadly, embedding link dynamics into richer general equilibrium environments with capital accumulation, multi-product firms, or international trade may shed light on how production networks adjust to structural change and macroeconomic shocks. By integrating high-quality data with theoretically grounded dynamics, this research contributes to a growing literature on the micro origins of aggregate productivity and the macroeconomic relevance of supply chain relationships.

References

- Acemoglu, Daron, Vasco M Carvalho, Asuman Ozdaglar, and Alireza Tahbaz-Salehi,** “The network origins of aggregate fluctuations,” *Econometrica*, 2012, 80 (5), 1977–2016.
- Baqae, David Rezza and Emmanuel Farhi,** “The macroeconomic impact of microeconomic shocks: Beyond Hulten’s theorem,” *Econometrica*, 2019, 87 (4), 1155–1203.
- Bartelsman, Eric, John Haltiwanger, and Stefano Scarpetta,** “Measuring and analyzing cross-country differences in firm dynamics,” in “Producer dynamics: New evidence from micro data,” University of Chicago Press, 2009, pp. 15–76.
- Bernard, Andrew B, Emmanuel Dhyne, Glenn Magerman, Kalina Manova, and Andreas Moxnes,** “The origins of firm heterogeneity: A production network approach,” *Journal of Political Economy*, 2022, 130 (7), 1765–1804.
- Carvalho, Vasco M,** “From micro to macro via production networks,” *Journal of Economic Perspectives*, 2014, 28 (4), 23–48.
- Davis, Steven J and John Haltiwanger,** “Gross job creation and destruction: Microeconomic evidence and macroeconomic implications,” *NBER macroeconomics annual*, 1990, 5, 123–168.
- Dhyne, Emmanuel, Ayumu Ken Kikkawa, Magne Mogstad, and Felix Tintelnot,** “Trade and domestic production networks,” *The Review of Economic Studies*, 2021, 88 (2), 643–668.
- Hsieh, Chang-Tai and Peter J Klenow,** “Misallocation and manufacturing TFP in China and India,” *The Quarterly journal of economics*, 2009, 124 (4), 1403–1448.
- Koike-Mori, Yasutaka and Koki Okumura,** “Production Networks and R&D Misallocation,” *unpublished, UCLA*, 2024.
- Lim, Kevin,** “Endogenous Production Networks and the Business Cycle,” in “in” 2018.
- Loecker, Jan De and Chad Syverson,** “An industrial organization perspective on productivity,” in “Handbook of industrial organization,” Vol. 4, Elsevier, 2021, pp. 141–223.
- Monarch, Ryan and Tim Schmidt-Eisenlohr,** “Longevity and the value of trade relationships,” *Journal of International Economics*, 2023, 145, 103842.
- Oberfield, Ezra,** “A theory of input–output architecture,” *Econometrica*, 2018, 86 (2), 559–589.

A Numerical Decomposition of Welfare Changes

This appendix describes how we decompose finite welfare changes using Proposition 1, Lemma 1, and Corollary 1. The procedure distinguishes the contributions of productivity, production labor, link expansion, and match-quality composition. Because Proposition 1 is a differential identity, we evaluate its components along a sequence of equilibria connecting the baseline and shocked economies. This allows propagation weights and sourcing shares to adjust with the shock.

Counterfactual equilibria

Let δ denote the full log-productivity shock and let $v(z)$ indicate the firms targeted by the experiment. We index firms by their baseline types and define

$$\log z_t(z) = \log z + t\delta v(z), \quad t \in [0, 1]. \quad (37)$$

The targeted set is selected in the baseline and remains fixed. The measure G continues to assign the same mass to each group of firms; it is not recomputed by evaluating the baseline density at shocked productivities. All other primitives, including the distribution of match quality, are held fixed. A proportional productivity increase of g corresponds to $\delta = \log(1 + g)$.

At each t , we solve the decentralized endogenous-network equilibrium. Prices, sales, relationship cutoffs, and labor allocation therefore adjust jointly. The parameter t indexes the magnitude of the counterfactual shock, not calendar time. The resulting sequence is a numerical construction for decomposing a comparative-static welfare change.

We assume that the equilibria belong to a differentiable branch with positive production labor and positive bilateral sourcing capacities. This is a local regularity requirement and does not establish global uniqueness of equilibrium. Throughout, z indexes buyers and z' indexes suppliers in bilateral objects.

The welfare channels in Proposition 1

With wages as the numeraire, Proposition 1 gives

$$d \log W = d \log L^Y + \tilde{\Lambda}^T \left[d \log z + \frac{1 - \alpha}{\sigma - 1} d \log Q \right], \quad (38)$$

where $\tilde{\Lambda}$ denotes the cost-based propagation weights and the sourcing-capacity differential satisfies

$$d \log Q_t(z) = \int \omega_t(z, z') d \log q_t(z, z') dG(z'). \quad (39)$$

The sourcing shares $\omega_t(z, z')$ are evaluated at the equilibrium associated with t and satisfy $\int \omega_t(z, z') dG(z') = 1$.

Integrating equation (38) gives

$$\log W_1 - \log W_0 = C_L + C_z + C_Q, \quad (40)$$

where

$$C_L = \log L_1^Y - \log L_0^Y, \quad (41)$$

$$C_z = \int_0^1 \int \tilde{\Lambda}_t(z) \delta v(z) dG(z) dt, \quad (42)$$

$$C_Q = \frac{1-\alpha}{\sigma-1} \int_0^1 \int \int \tilde{\Lambda}_t(z) \omega_t(z, z') \frac{\partial \log q_t(z, z')}{\partial t} dG(z') dG(z) dt. \quad (43)$$

The labor contribution measures the welfare effect of changes in labor available for production as relationship maintenance adjusts. The productivity contribution values the exogenous productivity changes using the propagation weights prevailing at each equilibrium. The sourcing contribution measures the effect of changes in bilateral productive capacity, accounting for both sourcing shares and propagation through the network.

Link expansion and match-quality composition

Lemma 1 decomposes bilateral sourcing capacity as

$$q_t(z, z') = \ell_t(z, z') \bar{r}_t(z, z'), \quad (44)$$

where

$$\ell_t(z, z') = \Pr(r \geq r_t^*(z, z')), \quad (45)$$

$$\bar{r}_t(z, z') = \mathbb{E} \left[r^{\sigma-1} \mid r \geq r_t^*(z, z') \right]. \quad (46)$$

Here $\bar{r}$ is the average productive intensity of an active relationship, rather than its average quality level.

Taking logarithms yields the exact identity

$$d \log q_t(z, z') = d \log \ell_t(z, z') + d \log \bar{r}_t(z, z'). \quad (47)$$

Substituting into equation (43) gives

$$C_Q = C_\ell + C_{\bar{r}}, \quad (48)$$

with

$$C_\ell = \frac{1-\alpha}{\sigma-1} \int_0^1 \int \int \tilde{\lambda}_t(z) \omega_t(z, z') \frac{\partial \log \ell_t(z, z')}{\partial t} dG(z') dG(z) dt, \quad (49)$$

$$C_{\bar{r}} = \frac{1-\alpha}{\sigma-1} \int_0^1 \int \int \tilde{\lambda}_t(z) \omega_t(z, z') \frac{\partial \log \bar{r}_t(z, z')}{\partial t} dG(z') dG(z) dt. \quad (50)$$

Consequently, the welfare change admits the four-channel decomposition

$$\boxed{\log W_1 - \log W_0 = C_L + C_z + C_\ell + C_{\bar{r}}.} \quad (51)$$

The link-expansion contribution records changes in the probability of an active relationship. The composition contribution records changes in average productive intensity among active relationships. These components describe jointly determined selection responses; they are not separate counterfactual experiments.

The role of the marginal relationship

Corollary 1 sharpens the interpretation of this decomposition. Holding the quality distribution H fixed, define

$$\theta_t(z, z') = \frac{r_t^*(z, z')^{\sigma-1}}{\bar{r}_t(z, z')}. \quad (52)$$

Assume that H has a continuous density h , that the cutoff is positive, and that the relevant probabilities and moments are positive and finite. Differentiating the cutoff integrals gives

$$d\ell = -h(r^*) dr^*, \quad dq = -(r^*)^{\sigma-1} h(r^*) dr^*. \quad (53)$$

Using $q = \ell \bar{r}$ therefore yields

$$d \log q = \theta d \log \ell, \quad (54)$$

$$d \log \bar{r} = -(1 - \theta) d \log \ell. \quad (55)$$

For $\sigma > 1$, active relationships have productive intensity at least as large as that of the marginal relationship, so $0 < \theta \leq 1$.

A reduction in the cutoff expands the active set but admits relationships with lower productive intensity than the existing average. Link expansion therefore increases sourcing capacity, while composition offsets part of that increase. The ratio θ measures the fraction of the proportional link expansion that translates into a proportional capacity increase. Conversely, raising the cutoff reduces link probability while increasing average productive intensity among surviving relationships.

The corollary provides equivalent expressions for the integrated contributions:

$$C_Q = \frac{1-\alpha}{\sigma-1} \int_0^1 \int \int \tilde{\Lambda}_t(z) \omega_t(z, z') \theta_t(z, z') \frac{\partial \log \ell_t(z, z')}{\partial t} dG(z') dG(z) dt, \quad (56)$$

$$C_{\bar{r}} = -\frac{1-\alpha}{\sigma-1} \int_0^1 \int \int \tilde{\Lambda}_t(z) \omega_t(z, z') [1 - \theta_t(z, z')] \frac{\partial \log \ell_t(z, z')}{\partial t} dG(z') dG(z) dt. \quad (57)$$

These expressions also provide a numerical check on the decomposition obtained directly from Lemma 1. Aggregate contributions depend on the pattern of cutoff changes and their welfare weights; their signs need not follow the response of any particular type pair.

Numerical implementation

We divide the full shock into K equal increments and solve equilibria at $t_k = k/K$, for $k = 0, \dots, K$. Consecutive productivity vectors differ by

$$\Delta_k \log z(z) = \frac{\delta}{K} v(z). \quad (58)$$

Each equilibrium is initialized using the preceding solution. All solutions must satisfy the equilibrium convergence criteria before entering the decomposition.

Define the combined bilateral welfare weight

$$A_k(z, z') = \frac{1-\alpha}{\sigma-1} \tilde{\Lambda}_{t_k}(z) \omega_{t_k}(z, z'). \quad (59)$$

For any positive equilibrium object x , write $\Delta_k \log x = \log x_{t_{k+1}} - \log x_{t_k}$. We approxi-

mate the integrated contributions using endpoint averages:

$$\widehat{C}_z = \sum_{k=0}^{K-1} \int \frac{\widetilde{\Lambda}_{t_k}(z) + \widetilde{\Lambda}_{t_{k+1}}(z)}{2} \Delta_k \log z(z) dG(z), \quad (60)$$

$$\widehat{C}_\ell = \sum_{k=0}^{K-1} \iint \frac{A_k(z, z') + A_{k+1}(z, z')}{2} \Delta_k \log \ell(z, z') dG(z') dG(z), \quad (61)$$

$$\widehat{C}_{\bar{r}} = \sum_{k=0}^{K-1} \iint \frac{A_k(z, z') + A_{k+1}(z, z')}{2} \Delta_k \log \bar{r}(z, z') dG(z') dG(z). \quad (62)$$

The labor contribution is computed directly from the endpoints, $C_L = \log L_1^Y - \log L_0^Y$.

Because the factorization $q = \ell \bar{r}$ also implies

$$\Delta_k \log q = \Delta_k \log \ell + \Delta_k \log \bar{r}, \quad (63)$$

the two numerical sourcing components sum to the sourcing contribution computed from $\Delta_k \log q$, up to numerical precision, when identical weights are used.

For the corollary-based check, we replace the integrand weights by endpoint averages of the products $A\theta$ and $A(1 - \theta)$, respectively. These alternative calculations need not coincide exactly with the Lemma 1 calculations at finite K , because the corollary is a differential identity. They should converge to the same contributions as the integration grid is refined.

In the lognormal specification, link probabilities and productive capacities are evaluated using analytical cutoff integrals. This avoids the artificial jumps in selection generated by a finite quality grid. Baseline, intermediate, and final equilibria use the same integration method.

We assess accuracy by increasing K and reporting the accounting residual

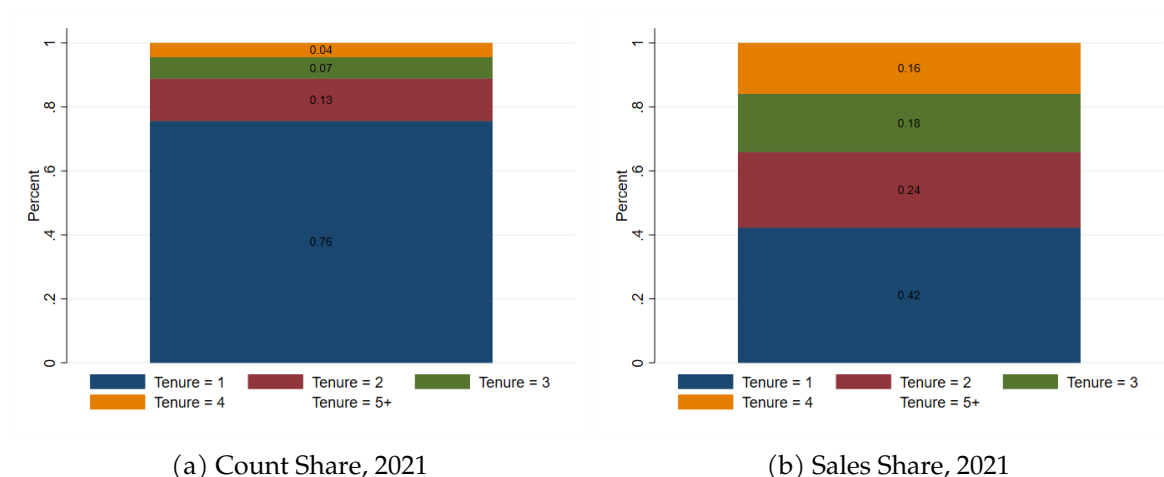
$$R_K = \log W_1 - \log W_0 - [C_L + \widehat{C}_z + \widehat{C}_\ell + \widehat{C}_{\bar{r}}]. \quad (64)$$

Under the stated regularity conditions, this residual should approach zero as numerical integration and equilibrium solutions become sufficiently accurate.

Interpretation. The decomposition is additive in log welfare units. Its components are evaluated along the proportional log-productivity experiment in equation (37); individual contributions can depend on the specified counterfactual sequence, although their exact sum equals the endpoint welfare change.

In particular, C_z uses propagation weights that adjust under endogenous network

Figure A1: Distribution of Links with observed birth



Note:

selection. It need not equal the finite welfare change from a separately solved fixed-network counterfactual. The decomposition and the fixed-network comparison therefore answer complementary questions: the former accounts for the channels operating within the endogenous equilibrium response, while the latter measures how the total response changes when network adjustment is restricted.

B Robustness

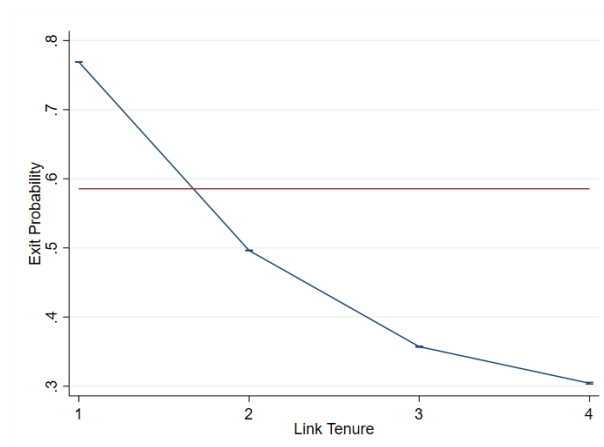
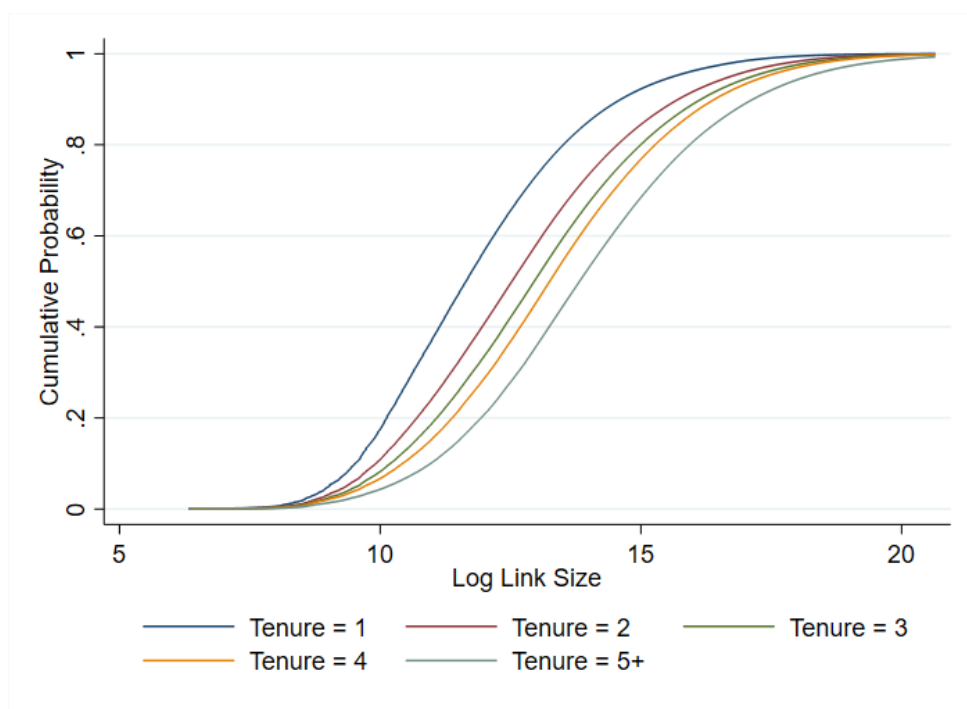


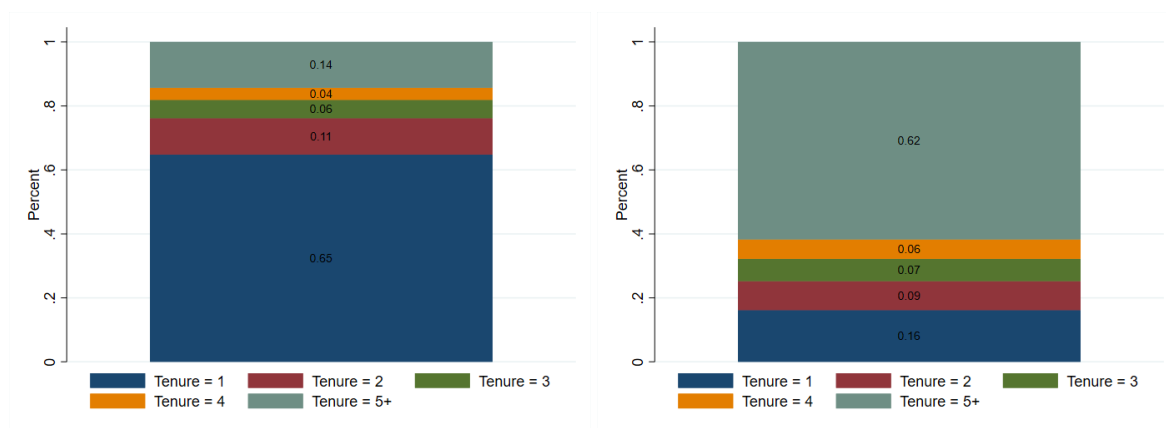
Figure A5: Robustness (Partial-Year): Hazard and Turnover by Link Tenure

Figure A2: Link Tenure and FOSD



Note:

Figure A3: Robustness (Partial-Year): Count Share by Link Tenure



(a) Count Share, 2021

(b) Sales Share, 2021

Note:

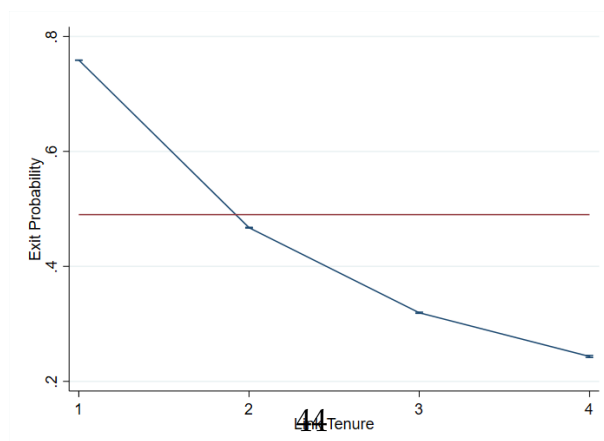
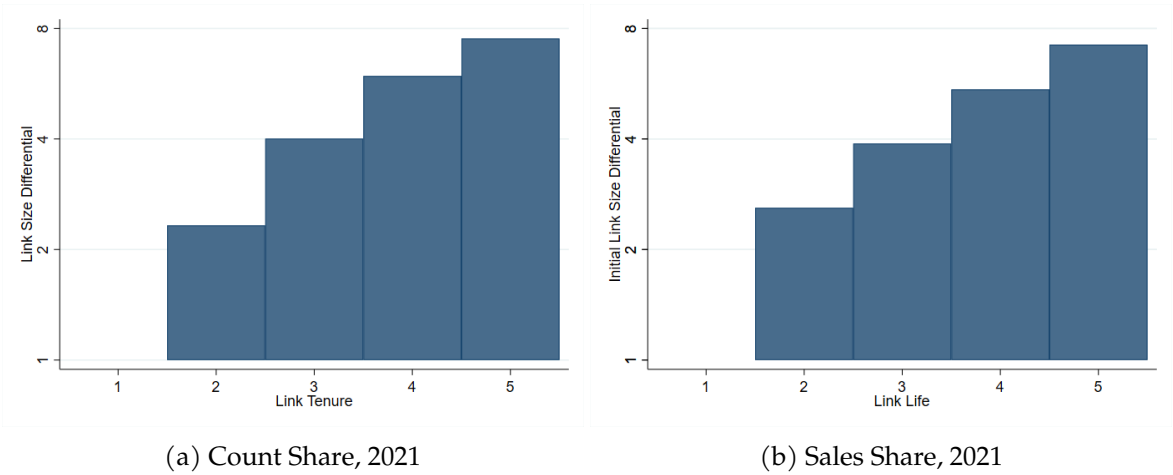


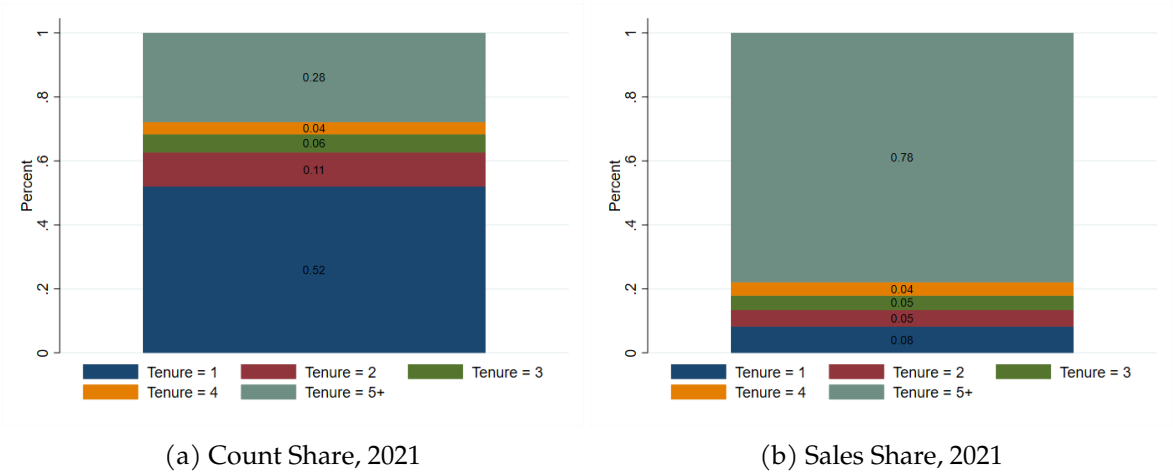
Figure A8: Robustness (Balanced Panel): Hazard and Turnover by Link Tenure

Figure A4: Robustness (Partial-Year): Link Value by Link Tenure



Note:

Figure A6: Robustness (Balanced Panel): Count Share by Link Tenure



Note:

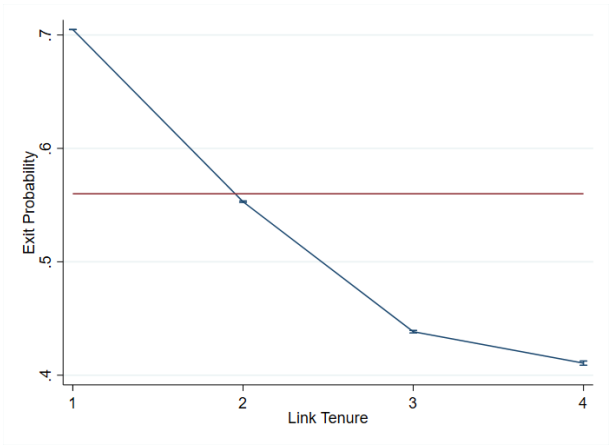
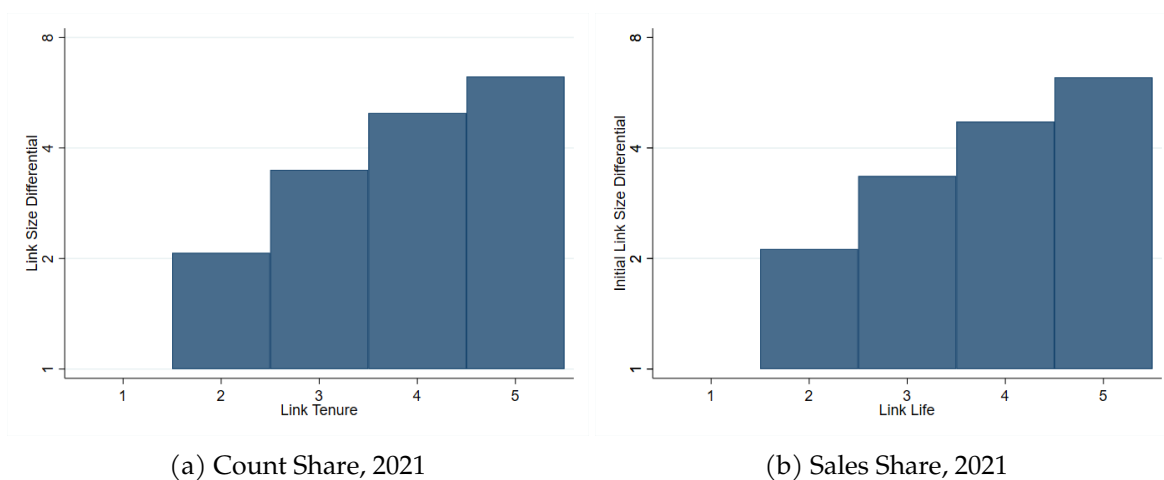


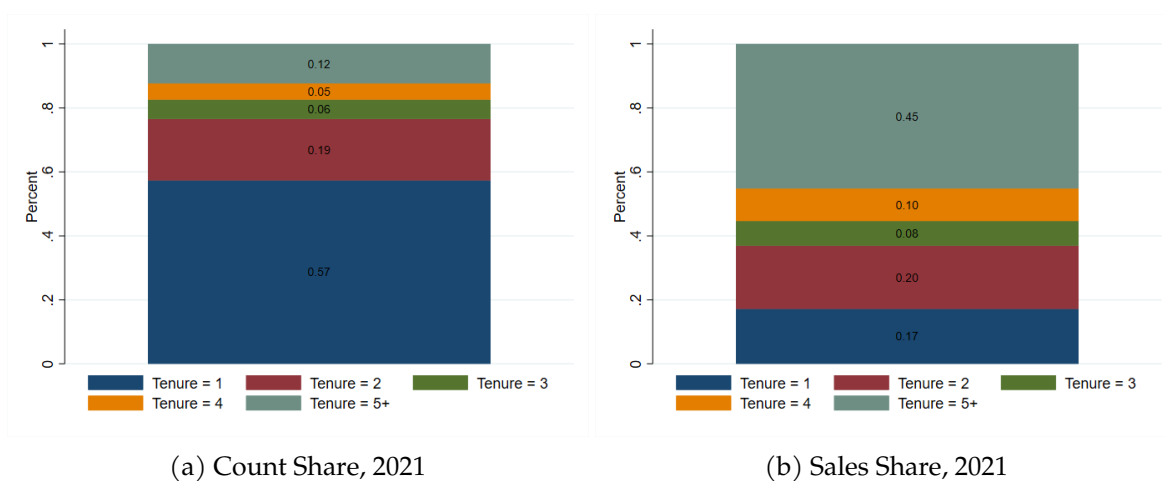
Figure A11: Robustness (Excluding Retail/Wholesale): Hazard and Turnover by Link Tenure

Figure A7: Robustness (Balanced Panel): Link Value by Link Tenure



Note:

Figure A9: Robustness (Excluding Retail/Wholesale): Count Share by Link Tenure



Note:

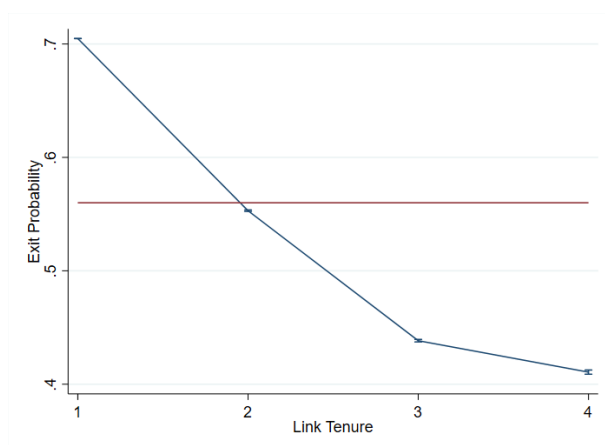
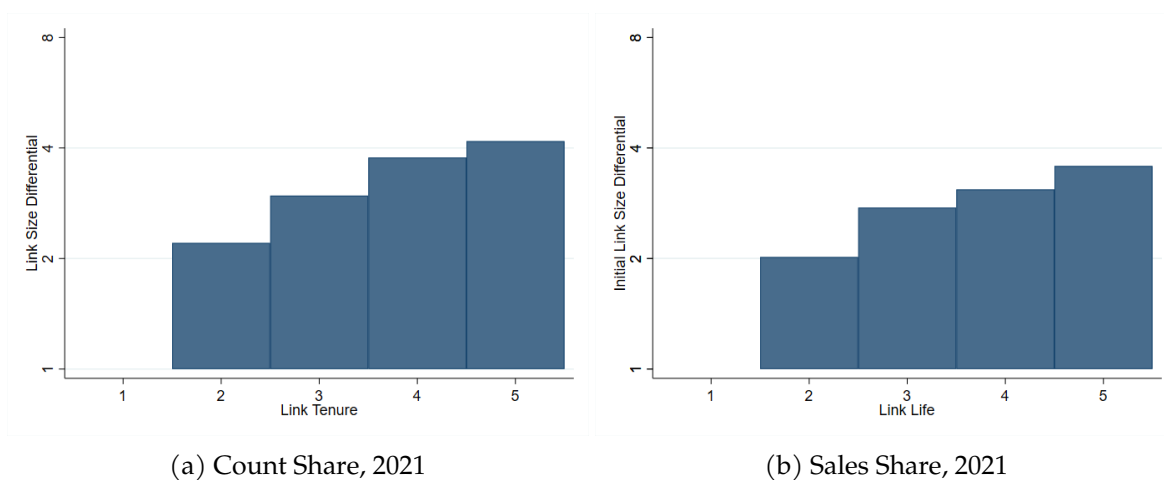


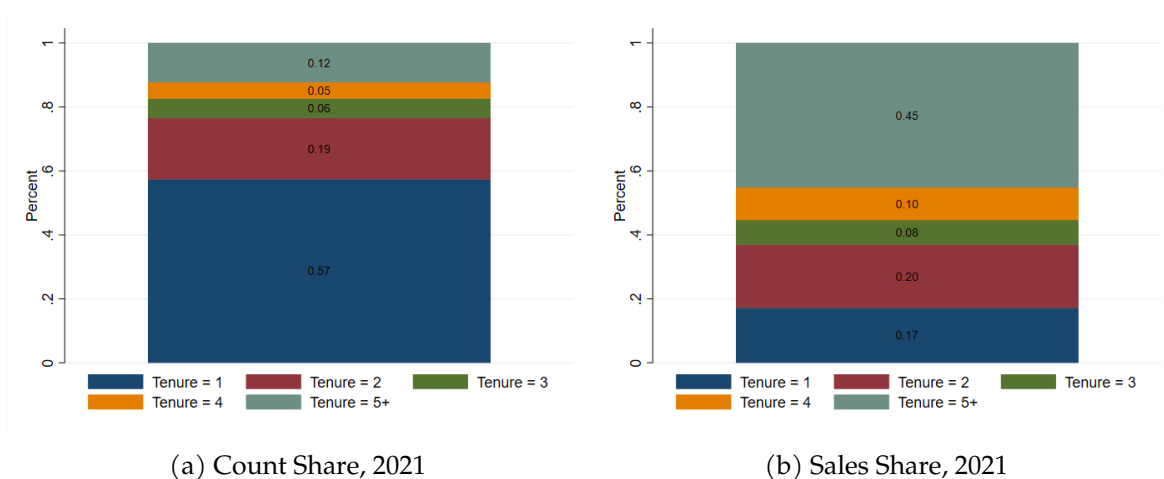
Figure A14: Robustness (Excluding Capital-Intensive Sectors): Hazard and Turnover by Link Tenure

Figure A10: Robustness (Excluding Retail/Wholesale): Link Value by Link Tenure



Note:

Figure A12: Robustness (Excluding Capital-Intensive Sectors): Count Share by Link Tenure



Note:

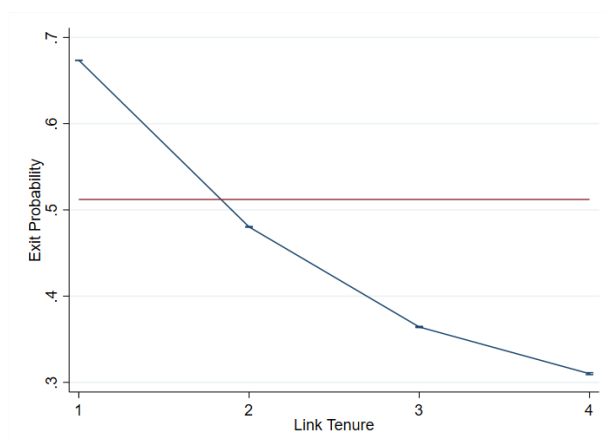
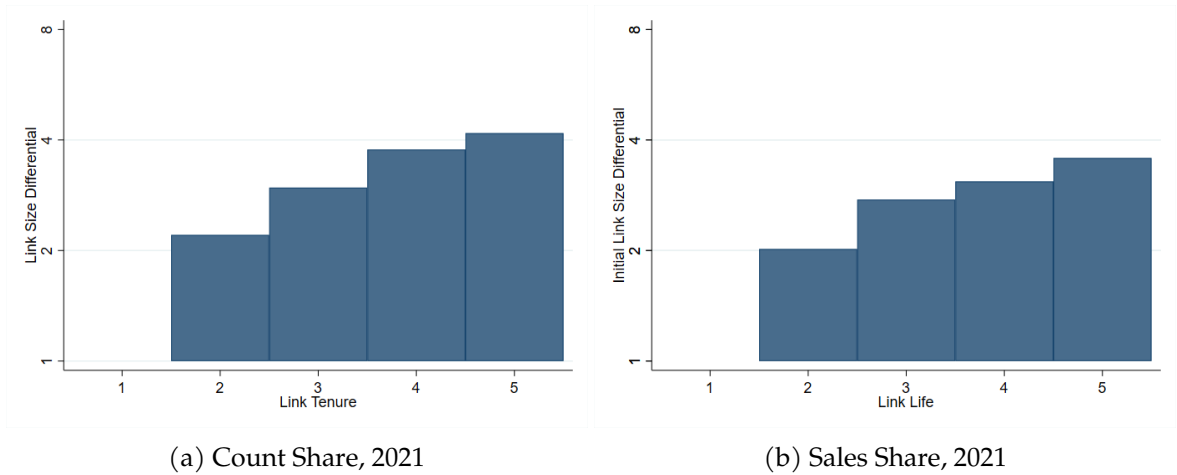


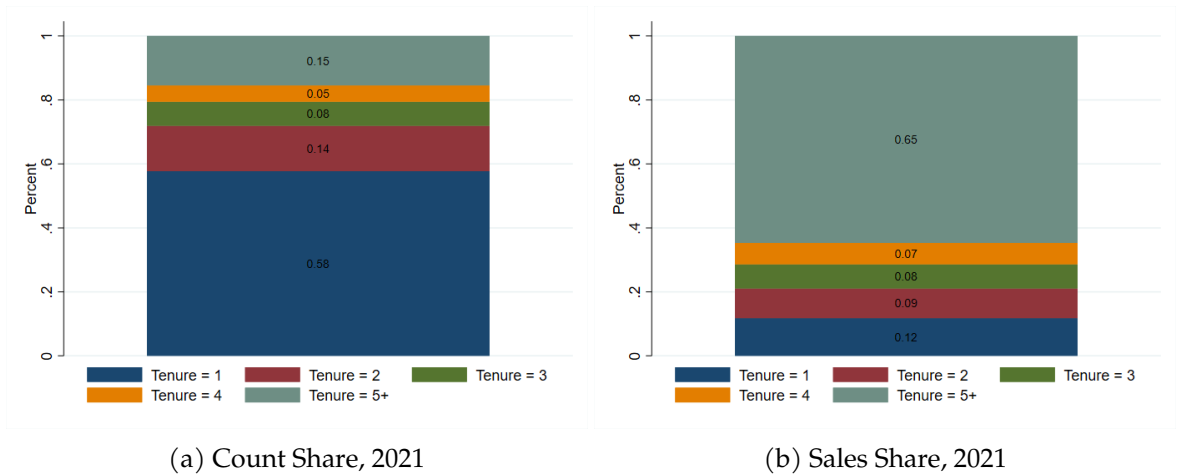
Figure A17: Robustness (Cross Section): Hazard and Turnover by Link Tenure

Figure A13: Robustness (Excluding Capital-Intensive Sectors): Link Value by Link Tenure



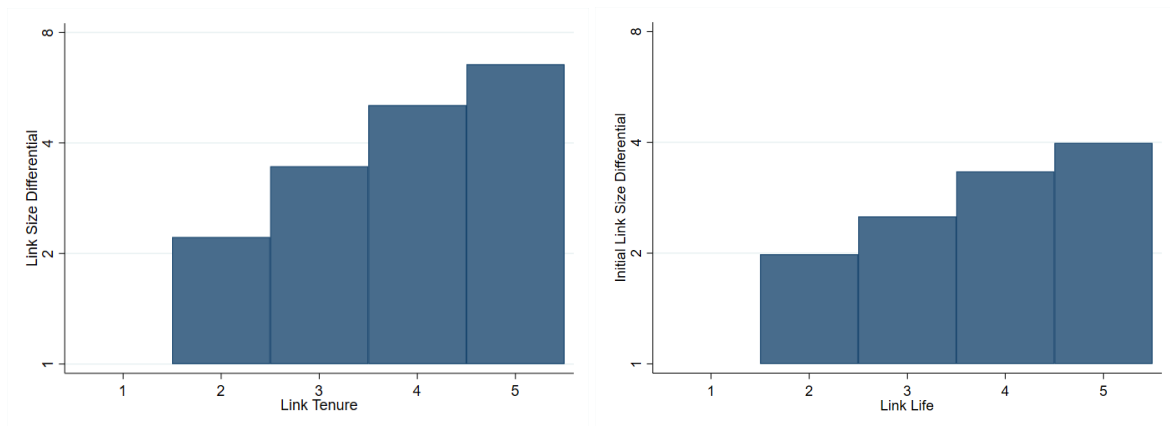
Note:

Figure A15: Robustness (Cross Section): Count Share by Link Tenure



Note:

Figure A16: Robustness (Cross Section): Link Value by Link Tenure



(a) Count Share, 2021

(b) Sales Share, 2021

Note: